Chargedoff_accounts_analysis

October 25, 2026

[1]:
```
# %pip install plotly
```

[2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    roc_auc_score,
    average_precision_score,
    brier_score_loss,
    precision_score,
    recall_score,
    f1_score
)
```

[3]:
```
pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", None)
```

[4]:
```
df=pd.read_csv("loan.csv")
```

```
/var/folders/gl/xrbhkv5j2rb0pw50rbz6n8lc0000gn/T/ipykernel_27423/1218720392.py:1
: DtypeWarning: Columns (19,47,55,112,123,124,125,128,129,130,133,139,140,141)
have mixed types. Specify dtype option on import or set low_memory=False.
  df=pd.read_csv("loan.csv")
```

0.0.1 1. Data Understanding

[5]:
```
df.head()
```

[5]:

	id	member_id	loan_amnt	funded_amnt	funded_amnt_inv	term
0	NaN	NaN	2500	2500	2500.0	36 months
1	NaN	NaN	30000	30000	30000.0	60 months
2	NaN	NaN	5000	5000	5000.0	36 months
3	NaN	NaN	4000	4000	4000.0	36 months
4	NaN	NaN	30000	30000	30000.0	60 months

	int_rate	installment	grade	sub_grade	emp_title	emp_length
0	13.56	84.92	C	C1	Chef	10+ years
1	18.94	777.23	D	D2	Postmaster	10+ years
2	17.97	180.69	D	D1	Administrative	6 years
3	18.94	146.51	D	D2	IT Supervisor	10+ years
4	16.14	731.78	C	C4	Mechanic	10+ years

	home_ownership	annual_inc	verification_status	issue_d	loan_status
0	RENT	55000.0	Not Verified	Dec-2018	Current
1	MORTGAGE	90000.0	Source Verified	Dec-2018	Current
2	MORTGAGE	59280.0	Source Verified	Dec-2018	Current
3	MORTGAGE	92000.0	Source Verified	Dec-2018	Current
4	MORTGAGE	57250.0	Not Verified	Dec-2018	Current

	pymnt_plan	url	desc	purpose	title	zip_code
0	n	NaN	NaN	debt_consolidation	Debt consolidation	109xx
1	n	NaN	NaN	debt_consolidation	Debt consolidation	713xx
2	n	NaN	NaN	debt_consolidation	Debt consolidation	490xx
3	n	NaN	NaN	debt_consolidation	Debt consolidation	985xx
4	n	NaN	NaN	debt_consolidation	Debt consolidation	212xx

	addr_state	dti	delinq_2yrs	earliest_cr_line	inq_last_6mths
0	NY	18.24	0.0	Apr-2001	1.0
1	LA	26.52	0.0	Jun-1987	0.0
2	MI	10.51	0.0	Apr-2011	0.0
3	WA	16.74	0.0	Feb-2006	0.0
4	MD	26.35	0.0	Dec-2000	0.0

	mths_since_last_delinq	mths_since_last_record	open_acc	pub_rec
0	NaN	45.0	9.0	1.0
1	71.0	75.0	13.0	1.0
2	NaN	NaN	8.0	0.0
3	NaN	NaN	10.0	0.0
4	NaN	NaN	12.0	0.0

	revol_bal	revol_util	total_acc	initial_list_status	out_prncp
0	4341	10.3	34.0	w	2386.02
1	12315	24.2	44.0	w	29387.75
2	4599	19.1	13.0	w	4787.21
3	5468	78.1	13.0	w	3831.93

```
4        829         3.6       26.0                   w   29339.02

   out_prncp_inv  total_pymnt  total_pymnt_inv  total_rec_prncp  \
0        2386.02       167.02           167.02           113.98
1       29387.75      1507.11          1507.11           612.25
2        4787.21       353.89           353.89           212.79
3        3831.93       286.71           286.71           168.07
4       29339.02      1423.21          1423.21           660.98

   total_rec_int  total_rec_late_fee  recoveries  collection_recovery_fee  \
0          53.04                 0.0         0.0                      0.0
1         894.86                 0.0         0.0                      0.0
2         141.10                 0.0         0.0                      0.0
3         118.64                 0.0         0.0                      0.0
4         762.23                 0.0         0.0                      0.0

  last_pymnt_d  last_pymnt_amnt next_pymnt_d last_credit_pull_d  \
0    Feb-2019            84.92     Mar-2019          Feb-2019
1    Feb-2019           777.23     Mar-2019          Feb-2019
2    Feb-2019           180.69     Mar-2019          Feb-2019
3    Feb-2019           146.51     Mar-2019          Feb-2019
4    Feb-2019           731.78     Mar-2019          Feb-2019

   collections_12_mths_ex_med  mths_since_last_major_derog  policy_code  \
0                         0.0                          NaN            1
1                         0.0                          NaN            1
2                         0.0                          NaN            1
3                         0.0                          NaN            1
4                         0.0                          NaN            1

  application_type  annual_inc_joint  dti_joint verification_status_joint  \
0       Individual               NaN        NaN                       NaN
1       Individual               NaN        NaN                       NaN
2       Individual               NaN        NaN                       NaN
3       Individual               NaN        NaN                       NaN
4       Individual               NaN        NaN                       NaN

   acc_now_delinq  tot_coll_amt  tot_cur_bal  open_acc_6m  open_act_il  \
0             0.0           0.0      16901.0          2.0          2.0
1             0.0        1208.0     321915.0          4.0          4.0
2             0.0           0.0     110299.0          0.0          1.0
3             0.0         686.0     305049.0          1.0          5.0
4             0.0           0.0     116007.0          3.0          5.0

   open_il_12m  open_il_24m  mths_since_rcnt_il  total_bal_il  il_util  \
0          1.0          2.0                 2.0       12560.0     69.0
1          2.0          3.0                 3.0       87153.0     88.0
```

```
2          0.0          2.0                14.0        7150.0     72.0
3          3.0          5.0                 5.0       30683.0     68.0
4          3.0          5.0                 4.0       28845.0     89.0

   open_rv_12m  open_rv_24m  max_bal_bc  all_util  total_rev_hi_lim  inq_fi  \
0          2.0          7.0      2137.0      28.0           42000.0     1.0
1          4.0          5.0       998.0      57.0           50800.0     2.0
2          0.0          2.0         0.0      35.0           24100.0     1.0
3          0.0          0.0      3761.0      70.0            7000.0     2.0
4          2.0          4.0       516.0      54.0           23100.0     1.0

   total_cu_tl  inq_last_12m  acc_open_past_24mths  avg_cur_bal  \
0         11.0           2.0                   9.0       1878.0
1         15.0           2.0                  10.0      24763.0
2          5.0           0.0                   4.0      18383.0
3          4.0           3.0                   5.0      30505.0
4          0.0           0.0                   9.0       9667.0

   bc_open_to_buy  bc_util  chargeoff_within_12_mths  delinq_amnt  \
0         34360.0      5.9                       0.0          0.0
1         13761.0      8.3                       0.0          0.0
2         13800.0      0.0                       0.0          0.0
3          1239.0     75.2                       0.0          0.0
4          8471.0      8.9                       0.0          0.0

   mo_sin_old_il_acct  mo_sin_old_rev_tl_op  mo_sin_rcnt_rev_tl_op  \
0               140.0                 212.0                    1.0
1               163.0                 378.0                    4.0
2                87.0                  92.0                   15.0
3                62.0                 154.0                   64.0
4                53.0                 216.0                    2.0

   mo_sin_rcnt_tl  mort_acc  mths_since_recent_bc  mths_since_recent_bc_dlq  \
0             1.0       0.0                   1.0                       NaN
1             3.0       3.0                   4.0                       NaN
2            14.0       2.0                  77.0                       NaN
3             5.0       3.0                  64.0                       NaN
4             2.0       2.0                   2.0                       NaN

   mths_since_recent_inq  mths_since_recent_revol_delinq  \
0                    2.0                             NaN
1                    4.0                             NaN
2                   14.0                             NaN
3                    5.0                             NaN
4                   13.0                             NaN

   num_accts_ever_120_pd  num_actv_bc_tl  num_actv_rev_tl  num_bc_sats  \
```

```
0                   0.0             2.0             5.0          3.0
1                   0.0             2.0             4.0          4.0
2                   0.0             0.0             3.0          3.0
3                   0.0             1.0             2.0          1.0
4                   0.0             2.0             2.0          3.0

   num_bc_tl  num_il_tl  num_op_rev_tl  num_rev_accts  num_rev_tl_bal_gt_0  \
0        3.0       16.0            7.0           18.0                  5.0
1        9.0       27.0            8.0           14.0                  4.0
2        3.0        4.0            6.0            7.0                  3.0
3        2.0        7.0            2.0            3.0                  2.0
4        8.0        9.0            6.0           15.0                  2.0

   num_sats  num_tl_120dpd_2m  num_tl_30dpd  num_tl_90g_dpd_24m  \
0       9.0               0.0           0.0                 0.0
1      13.0               0.0           0.0                 0.0
2       8.0               0.0           0.0                 0.0
3      10.0               0.0           0.0                 0.0
4      12.0               0.0           0.0                 0.0

   num_tl_op_past_12m  pct_tl_nvr_dlq  percent_bc_gt_75  pub_rec_bankruptcies  \
0                 3.0           100.0               0.0                   1.0
1                 6.0            95.0               0.0                   1.0
2                 0.0           100.0               0.0                   0.0
3                 3.0           100.0             100.0                   0.0
4                 5.0            92.3               0.0                   0.0

   tax_liens  tot_hi_cred_lim  total_bal_ex_mort  total_bc_limit  \
0        0.0          60124.0            16901.0         36500.0
1        0.0         372872.0            99468.0         15000.0
2        0.0         136927.0            11749.0         13800.0
3        0.0         385183.0            36151.0          5000.0
4        0.0         157548.0            29674.0          9300.0

   total_il_high_credit_limit  revol_bal_joint sec_app_earliest_cr_line  \
0                     18124.0              NaN                      NaN
1                     94072.0              NaN                      NaN
2                     10000.0              NaN                      NaN
3                     44984.0              NaN                      NaN
4                     32332.0              NaN                      NaN

   sec_app_inq_last_6mths  sec_app_mort_acc  sec_app_open_acc  \
0                     NaN               NaN               NaN
1                     NaN               NaN               NaN
2                     NaN               NaN               NaN
3                     NaN               NaN               NaN
4                     NaN               NaN               NaN
```

```
   sec_app_revol_util  sec_app_open_act_il  sec_app_num_rev_accts  \
0                 NaN                  NaN                    NaN
1                 NaN                  NaN                    NaN
2                 NaN                  NaN                    NaN
3                 NaN                  NaN                    NaN
4                 NaN                  NaN                    NaN

   sec_app_chargeoff_within_12_mths  sec_app_collections_12_mths_ex_med  \
0                               NaN                                NaN
1                               NaN                                NaN
2                               NaN                                NaN
3                               NaN                                NaN
4                               NaN                                NaN

   sec_app_mths_since_last_major_derog hardship_flag hardship_type  \
0                                  NaN             N           NaN
1                                  NaN             N           NaN
2                                  NaN             N           NaN
3                                  NaN             N           NaN
4                                  NaN             N           NaN

  hardship_reason hardship_status  deferral_term  hardship_amount  \
0             NaN             NaN            NaN              NaN
1             NaN             NaN            NaN              NaN
2             NaN             NaN            NaN              NaN
3             NaN             NaN            NaN              NaN
4             NaN             NaN            NaN              NaN

  hardship_start_date hardship_end_date payment_plan_start_date  \
0                 NaN               NaN                    NaN
1                 NaN               NaN                    NaN
2                 NaN               NaN                    NaN
3                 NaN               NaN                    NaN
4                 NaN               NaN                    NaN

   hardship_length  hardship_dpd hardship_loan_status  \
0              NaN           NaN                  NaN
1              NaN           NaN                  NaN
2              NaN           NaN                  NaN
3              NaN           NaN                  NaN
4              NaN           NaN                  NaN

   orig_projected_additional_accrued_interest  hardship_payoff_balance_amount  \
0                                         NaN                             NaN
1                                         NaN                             NaN
2                                         NaN                             NaN
```

```
3                                        NaN                                 NaN
4                                        NaN                                 NaN

   hardship_last_payment_amount disbursement_method debt_settlement_flag  \
0                           NaN               Cash                     N
1                           NaN               Cash                     N
2                           NaN               Cash                     N
3                           NaN               Cash                     N
4                           NaN               Cash                     N

  debt_settlement_flag_date settlement_status settlement_date  \
0                       NaN               NaN             NaN
1                       NaN               NaN             NaN
2                       NaN               NaN             NaN
3                       NaN               NaN             NaN
4                       NaN               NaN             NaN

   settlement_amount  settlement_percentage  settlement_term
0                NaN                    NaN             NaN
1                NaN                    NaN             NaN
2                NaN                    NaN             NaN
3                NaN                    NaN             NaN
4                NaN                    NaN             NaN
```

[6]:
```
df.sample()
```

[6]:
```
        id  member_id  loan_amnt  funded_amnt  funded_amnt_inv        term  \
951901 NaN        NaN      15000        15000          15000.0   60 months

        int_rate  installment grade sub_grade        emp_title emp_length  \
951901      18.2       382.54     E        E1  Project manager    8 years

       home_ownership  annual_inc verification_status   issue_d  loan_status  \
951901       MORTGAGE    125000.0     Source Verified  Dec-2015  Charged Off

       pymnt_plan  url desc           purpose             title zip_code  \
951901          n  NaN  NaN  home_improvement  Home improvement    773xx

       addr_state   dti  delinq_2yrs earliest_cr_line  inq_last_6mths  \
951901         TX  5.03          0.0         Apr-2005             0.0

        mths_since_last_delinq  mths_since_last_record  open_acc  pub_rec  \
951901                     NaN                     NaN      10.0      0.0

        revol_bal  revol_util  total_acc initial_list_status  out_prncp  \
951901      26724        87.6       18.0                   w        0.0
```

```
        out_prncp_inv  total_pymnt  total_pymnt_inv  total_rec_prncp  \
951901            0.0      6970.27          6970.27          2398.08

        total_rec_int  total_rec_late_fee  recoveries  \
951901        2942.31                0.0     1629.88

        collection_recovery_fee last_pymnt_d  last_pymnt_amnt next_pymnt_d  \
951901                293.3784     Feb-2017           382.54          NaN

       last_credit_pull_d  collections_12_mths_ex_med  \
951901          Aug-2017                         0.0

        mths_since_last_major_derog  policy_code application_type  \
951901                          NaN            1       Individual

        annual_inc_joint  dti_joint verification_status_joint  acc_now_delinq  \
951901               NaN        NaN                       NaN             0.0

        tot_coll_amt  tot_cur_bal  open_acc_6m  open_act_il  open_il_12m  \
951901           0.0     254624.0          NaN          NaN          NaN

        open_il_24m  mths_since_rcnt_il  total_bal_il  il_util  open_rv_12m  \
951901          NaN                 NaN           NaN      NaN          NaN

        open_rv_24m  max_bal_bc  all_util  total_rev_hi_lim  inq_fi  \
951901          NaN         NaN       NaN           30500.0     NaN

        total_cu_tl  inq_last_12m  acc_open_past_24mths  avg_cur_bal  \
951901          NaN           NaN                   8.0      25462.0

        bc_open_to_buy  bc_util  chargeoff_within_12_mths  delinq_amnt  \
951901           187.0     98.3                       0.0          0.0

        mo_sin_old_il_acct  mo_sin_old_rev_tl_op  mo_sin_rcnt_rev_tl_op  \
951901               122.0                 128.0                    6.0

        mo_sin_rcnt_tl  mort_acc  mths_since_recent_bc  \
951901             6.0       2.0                  13.0

        mths_since_recent_bc_dlq  mths_since_recent_inq  \
951901                       NaN                    7.0

        mths_since_recent_revol_delinq  num_accts_ever_120_pd  num_actv_bc_tl  \
951901                             NaN                    0.0             3.0

        num_actv_rev_tl  num_bc_sats  num_bc_tl  num_il_tl  num_op_rev_tl  \
951901              7.0          3.0        7.0        4.0            8.0
```

```
        num_rev_accts  num_rev_tl_bal_gt_0  num_sats  num_tl_120dpd_2m  \
951901           12.0                  7.0      10.0               0.0

        num_tl_30dpd  num_tl_90g_dpd_24m  num_tl_op_past_12m  pct_tl_nvr_dlq  \
951901           0.0                 0.0                 5.0           100.0

        percent_bc_gt_75  pub_rec_bankruptcies  tax_liens  tot_hi_cred_lim  \
951901             100.0                   0.0        0.0         260750.0

        total_bal_ex_mort  total_bc_limit  total_il_high_credit_limit  \
951901            40974.0         10700.0                     14250.0

        revol_bal_joint sec_app_earliest_cr_line  sec_app_inq_last_6mths  \
951901              NaN                      NaN                     NaN

        sec_app_mort_acc  sec_app_open_acc  sec_app_revol_util  \
951901               NaN               NaN                 NaN

        sec_app_open_act_il  sec_app_num_rev_accts  \
951901                  NaN                    NaN

        sec_app_chargeoff_within_12_mths  sec_app_collections_12_mths_ex_med  \
951901                               NaN                                 NaN

        sec_app_mths_since_last_major_derog hardship_flag hardship_type  \
951901                                 NaN             N           NaN

       hardship_reason hardship_status  deferral_term  hardship_amount  \
951901             NaN             NaN            NaN              NaN

       hardship_start_date hardship_end_date payment_plan_start_date  \
951901                 NaN               NaN                     NaN

        hardship_length  hardship_dpd hardship_loan_status  \
951901              NaN           NaN                  NaN

        orig_projected_additional_accrued_interest  \
951901                                         NaN

        hardship_payoff_balance_amount  hardship_last_payment_amount  \
951901                             NaN                            NaN

       disbursement_method debt_settlement_flag debt_settlement_flag_date  \
951901                Cash                    N                      NaN

       settlement_status settlement_date  settlement_amount  \
```

```
951901               NaN             NaN                NaN

        settlement_percentage  settlement_term
951901                    NaN              NaN
```

[7]:
```
df.describe()
```

[7]:
```
              id  member_id     loan_amnt   funded_amnt  funded_amnt_inv  \
count  0.0        0.0       2.260668e+06  2.260668e+06     2.260668e+06
mean   NaN        NaN       1.504693e+04  1.504166e+04     1.502344e+04
std    NaN        NaN       9.190245e+03  9.188413e+03     9.192332e+03
min    NaN        NaN       5.000000e+02  5.000000e+02     0.000000e+00
25%    NaN        NaN       8.000000e+03  8.000000e+03     8.000000e+03
50%    NaN        NaN       1.290000e+04  1.287500e+04     1.280000e+04
75%    NaN        NaN       2.000000e+04  2.000000e+04     2.000000e+04
max    NaN        NaN       4.000000e+04  4.000000e+04     4.000000e+04

           int_rate   installment    annual_inc  url           dti  \
count  2.260668e+06  2.260668e+06  2.260664e+06  0.0  2.258957e+06
mean   1.309291e+01  4.458076e+02  7.799243e+04  NaN  1.882420e+01
std    4.832114e+00  2.671737e+02  1.126962e+05  NaN  1.418333e+01
min    5.310000e+00  4.930000e+00  0.000000e+00  NaN -1.000000e+00
25%    9.490000e+00  2.516500e+02  4.600000e+04  NaN  1.189000e+01
50%    1.262000e+01  3.779900e+02  6.500000e+04  NaN  1.784000e+01
75%    1.599000e+01  5.933200e+02  9.300000e+04  NaN  2.449000e+01
max    3.099000e+01  1.719830e+03  1.100000e+08  NaN  9.990000e+02

         delinq_2yrs  inq_last_6mths  mths_since_last_delinq  \
count  2.260639e+06    2.260638e+06            1.102166e+06
mean   3.068792e-01    5.768354e-01            3.454092e+01
std    8.672303e-01    8.859632e-01            2.190047e+01
min    0.000000e+00    0.000000e+00            0.000000e+00
25%    0.000000e+00    0.000000e+00            1.600000e+01
50%    0.000000e+00    0.000000e+00            3.100000e+01
75%    0.000000e+00    1.000000e+00            5.000000e+01
max    5.800000e+01    3.300000e+01            2.260000e+02

       mths_since_last_record      open_acc       pub_rec     revol_bal  \
count           359156.000000  2.260639e+06  2.260639e+06  2.260668e+06
mean                72.312842  1.161240e+01  1.975278e-01  1.665846e+04
std                 26.464094  5.640861e+00  5.705150e-01  2.294831e+04
min                  0.000000  0.000000e+00  0.000000e+00  0.000000e+00
25%                 55.000000  8.000000e+00  0.000000e+00  5.950000e+03
50%                 74.000000  1.100000e+01  0.000000e+00  1.132400e+04
75%                 92.000000  1.400000e+01  0.000000e+00  2.024600e+04
max                129.000000  1.010000e+02  8.600000e+01  2.904836e+06
```

```
         revol_util     total_acc     out_prncp  out_prncp_inv   total_pymnt  \
count  2.258866e+06  2.260639e+06  2.260668e+06   2.260668e+06  2.260668e+06
mean   5.033770e+01  2.416255e+01  4.446293e+03   4.445295e+03  1.182403e+04
std    2.471307e+01  1.198753e+01  7.547612e+03   7.546657e+03  9.889599e+03
min    0.000000e+00  1.000000e+00  0.000000e+00   0.000000e+00  0.000000e+00
25%    3.150000e+01  1.500000e+01  0.000000e+00   0.000000e+00  4.272580e+03
50%    5.030000e+01  2.200000e+01  0.000000e+00   0.000000e+00  9.060870e+03
75%    6.940000e+01  3.100000e+01  6.712632e+03   6.710320e+03  1.670797e+04
max    8.923000e+02  1.760000e+02  4.000000e+04   4.000000e+04  6.329688e+04

       total_pymnt_inv  total_rec_prncp  total_rec_int  total_rec_late_fee  \
count     2.260668e+06     2.260668e+06   2.260668e+06        2.260668e+06
mean      1.180594e+04     9.300142e+03   2.386352e+03        1.462469e+00
std       9.884835e+03     8.304886e+03   2.663086e+03        1.150210e+01
min       0.000000e+00     0.000000e+00   0.000000e+00       -9.500000e-09
25%       4.257730e+03     2.846180e+03   6.936100e+02        0.000000e+00
50%       9.043080e+03     6.823385e+03   1.485280e+03        0.000000e+00
75%       1.668257e+04     1.339750e+04   3.052220e+03        0.000000e+00
max       6.329688e+04     4.000000e+04   2.819250e+04        1.427250e+03

         recoveries  collection_recovery_fee  last_pymnt_amnt  \
count  2.260668e+06             2.260668e+06     2.260668e+06
mean   1.360740e+02             2.259328e+01     3.364015e+03
std    7.258317e+02             1.271114e+02     5.971757e+03
min    0.000000e+00             0.000000e+00     0.000000e+00
25%    0.000000e+00             0.000000e+00     3.086400e+02
50%    0.000000e+00             0.000000e+00     5.884700e+02
75%    0.000000e+00             0.000000e+00     3.534965e+03
max    3.985955e+04             7.174719e+03     4.219205e+04

       collections_12_mths_ex_med  mths_since_last_major_derog  policy_code  \
count                2.260523e+06                 580775.000000    2260668.0
mean                 1.814580e-02                     44.164220          1.0
std                  1.508131e-01                     21.533121          0.0
min                  0.000000e+00                      0.000000          1.0
25%                  0.000000e+00                     27.000000          1.0
50%                  0.000000e+00                     44.000000          1.0
75%                  0.000000e+00                     62.000000          1.0
max                  2.000000e+01                    226.000000          1.0

       annual_inc_joint      dti_joint  acc_now_delinq  tot_coll_amt  \
count      1.207100e+05  120706.000000    2.260639e+06  2.190392e+06
mean       1.236246e+05      19.251817    4.147942e-03  2.327317e+02
std        7.416135e+04       7.822086    6.961656e-02  8.518462e+03
min        5.693510e+03       0.000000    0.000000e+00  0.000000e+00
25%        8.340000e+04      13.530000    0.000000e+00  0.000000e+00
50%        1.100000e+05      18.840000    0.000000e+00  0.000000e+00
```

```
75%        1.479950e+05      24.620000    0.000000e+00  0.000000e+00
max        7.874821e+06      69.490000    1.400000e+01  9.152545e+06

        tot_cur_bal   open_acc_6m   open_act_il   open_il_12m   open_il_24m  \
count  2.190392e+06  1.394538e+06  1.394539e+06  1.394539e+06  1.394539e+06
mean   1.424922e+05  9.344199e-01  2.779407e+00  6.764314e-01  1.562752e+00
std    1.606926e+05  1.140700e+00  3.000784e+00  9.256354e-01  1.578672e+00
min    0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00
25%    2.909200e+04  0.000000e+00  1.000000e+00  0.000000e+00  0.000000e+00
50%    7.924000e+04  1.000000e+00  2.000000e+00  0.000000e+00  1.000000e+00
75%    2.132040e+05  1.000000e+00  3.000000e+00  1.000000e+00  2.000000e+00
max    9.971659e+06  1.800000e+01  5.700000e+01  2.500000e+01  5.100000e+01

       mths_since_rcnt_il  total_bal_il       il_util   open_rv_12m  \
count        1.350744e+06  1.394539e+06  1.191818e+06  1.394539e+06
mean         2.122236e+01  3.550665e+04  6.914098e+01  1.290133e+00
std          2.604919e+01  4.409746e+04  2.374839e+01  1.506827e+00
min          0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00
25%          7.000000e+00  8.695000e+03  5.500000e+01  0.000000e+00
50%          1.300000e+01  2.312700e+04  7.200000e+01  1.000000e+00
75%          2.400000e+01  4.609500e+04  8.600000e+01  2.000000e+00
max          5.110000e+02  1.837038e+06  1.000000e+03  2.800000e+01

        open_rv_24m    max_bal_bc      all_util  total_rev_hi_lim  \
count  1.394539e+06  1.394539e+06  1.394320e+06      2.190392e+06
mean   2.749923e+00  5.806393e+03  5.703230e+01      3.457394e+04
std    2.596911e+00  5.690561e+03  2.090475e+01      3.672850e+04
min    0.000000e+00  0.000000e+00  0.000000e+00      0.000000e+00
25%    1.000000e+00  2.284000e+03  4.300000e+01      1.470000e+04
50%    2.000000e+00  4.413000e+03  5.800000e+01      2.540000e+04
75%    4.000000e+00  7.598000e+03  7.200000e+01      4.320000e+04
max    6.000000e+01  1.170668e+06  2.390000e+02      9.999999e+06

             inq_fi   total_cu_tl  inq_last_12m  acc_open_past_24mths  \
count  1.394539e+06  1.394538e+06  1.394538e+06          2.210638e+06
mean   1.012867e+00  1.477304e+00  2.036667e+00          4.521656e+00
std    1.489456e+00  2.672991e+00  2.383117e+00          3.164229e+00
min    0.000000e+00  0.000000e+00  0.000000e+00          0.000000e+00
25%    0.000000e+00  0.000000e+00  0.000000e+00          2.000000e+00
50%    1.000000e+00  0.000000e+00  1.000000e+00          4.000000e+00
75%    1.000000e+00  2.000000e+00  3.000000e+00          6.000000e+00
max    4.800000e+01  1.110000e+02  6.700000e+01          6.400000e+01

        avg_cur_bal  bc_open_to_buy       bc_util  chargeoff_within_12_mths  \
count  2.190322e+06    2.185733e+06  2.184597e+06              2.260523e+06
mean   1.354780e+04    1.139426e+04  5.789995e+01              8.464413e-03
std    1.647408e+04    1.659953e+04  2.858347e+01              1.048098e-01
```

```
min    0.000000e+00    0.000000e+00  0.000000e+00              0.000000e+00
25%    3.080000e+03    1.722000e+03  3.540000e+01              0.000000e+00
50%    7.335000e+03    5.442000e+03  6.020000e+01              0.000000e+00
75%    1.878300e+04    1.418700e+04  8.310000e+01              0.000000e+00
max    9.580840e+05    7.111400e+05  3.396000e+02              1.000000e+01

        delinq_amnt  mo_sin_old_il_acct  mo_sin_old_rev_tl_op  \
count  2.260639e+06        2.121597e+06          2.190391e+06
mean   1.236983e+01        1.257378e+02          1.814916e+02
std    7.264648e+02        5.338218e+01          9.711845e+01
min    0.000000e+00        0.000000e+00          1.000000e+00
25%    0.000000e+00        9.600000e+01          1.160000e+02
50%    0.000000e+00        1.300000e+02          1.640000e+02
75%    0.000000e+00        1.540000e+02          2.320000e+02
max    2.499250e+05        9.990000e+02          9.990000e+02

       mo_sin_rcnt_rev_tl_op  mo_sin_rcnt_tl      mort_acc  \
count           2.190391e+06    2.190392e+06  2.210638e+06
mean            1.402409e+01    8.297469e+00  1.555382e+00
std             1.753308e+01    9.208557e+00  1.904981e+00
min             0.000000e+00    0.000000e+00  0.000000e+00
25%             4.000000e+00    3.000000e+00  0.000000e+00
50%             8.000000e+00    6.000000e+00  1.000000e+00
75%             1.700000e+01    1.100000e+01  3.000000e+00
max             5.470000e+02    3.820000e+02  9.400000e+01

       mths_since_recent_bc  mths_since_recent_bc_dlq  mths_since_recent_inq  \
count          2.187256e+06             519701.000000           1.965233e+06
mean           2.484485e+01                 39.303090           7.024194e+00
std            3.231925e+01                 22.617689           5.965411e+00
min            0.000000e+00                  0.000000           0.000000e+00
25%            6.000000e+00                 21.000000           2.000000e+00
50%            1.400000e+01                 37.000000           5.000000e+00
75%            3.000000e+01                 57.000000           1.100000e+01
max            6.610000e+02                202.000000           2.500000e+01

       mths_since_recent_revol_delinq  num_accts_ever_120_pd  num_actv_bc_tl  \
count                   740359.000000           2.190392e+06    2.190392e+06
mean                        35.782223           5.002082e-01    3.676069e+00
std                         22.307239           1.350326e+00    2.324646e+00
min                          0.000000           0.000000e+00    0.000000e+00
25%                         17.000000           0.000000e+00    2.000000e+00
50%                         33.000000           0.000000e+00    3.000000e+00
75%                         51.000000           0.000000e+00    5.000000e+00
max                        202.000000           5.800000e+01    5.000000e+01

       num_actv_rev_tl   num_bc_sats     num_bc_tl     num_il_tl  \
```

```
count     2.190392e+06  2.202078e+06  2.190392e+06  2.190392e+06
mean      5.629468e+00  4.774183e+00  7.726402e+00  8.413439e+00
std       3.382874e+00  3.037921e+00  4.701430e+00  7.359114e+00
min       0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00
25%       3.000000e+00  3.000000e+00  4.000000e+00  3.000000e+00
50%       5.000000e+00  4.000000e+00  7.000000e+00  6.000000e+00
75%       7.000000e+00  6.000000e+00  1.000000e+01  1.100000e+01
max       7.200000e+01  7.100000e+01  8.600000e+01  1.590000e+02

       num_op_rev_tl  num_rev_accts  num_rev_tl_bal_gt_0      num_sats  \
count   2.190392e+06   2.190391e+06         2.190392e+06  2.202078e+06
mean    8.246523e+00   1.400463e+01         5.577951e+00  1.162813e+01
std     4.683928e+00   8.038868e+00         3.293434e+00  5.644027e+00
min     0.000000e+00   0.000000e+00         0.000000e+00  0.000000e+00
25%     5.000000e+00   8.000000e+00         3.000000e+00  8.000000e+00
50%     7.000000e+00   1.200000e+01         5.000000e+00  1.100000e+01
75%     1.000000e+01   1.800000e+01         7.000000e+00  1.400000e+01
max     9.100000e+01   1.510000e+02         6.500000e+01  1.010000e+02

       num_tl_120dpd_2m  num_tl_30dpd  num_tl_90g_dpd_24m  num_tl_op_past_12m  \
count      2.107011e+06  2.190392e+06        2.190392e+06        2.190392e+06
mean       6.373958e-04  2.813652e-03        8.293767e-02        2.076755e+00
std        2.710643e-02  5.616522e-02        4.935732e-01        1.830711e+00
min        0.000000e+00  0.000000e+00        0.000000e+00        0.000000e+00
25%        0.000000e+00  0.000000e+00        0.000000e+00        1.000000e+00
50%        0.000000e+00  0.000000e+00        0.000000e+00        2.000000e+00
75%        0.000000e+00  0.000000e+00        0.000000e+00        3.000000e+00
max        7.000000e+00  4.000000e+00        5.800000e+01        3.200000e+01

       pct_tl_nvr_dlq  percent_bc_gt_75  pub_rec_bankruptcies     tax_liens  \
count    2.190237e+06      2.185289e+06          2.259303e+06  2.260563e+06
mean     9.411458e+01      4.243513e+01          1.281935e-01  4.677109e-02
std      9.036140e+00      3.621616e+01          3.646130e-01  3.775338e-01
min      0.000000e+00      0.000000e+00          0.000000e+00  0.000000e+00
25%      9.130000e+01      0.000000e+00          0.000000e+00  0.000000e+00
50%      1.000000e+02      3.750000e+01          0.000000e+00  0.000000e+00
75%      1.000000e+02      7.140000e+01          0.000000e+00  0.000000e+00
max      1.000000e+02      1.000000e+02          1.200000e+01  8.500000e+01

       tot_hi_cred_lim  total_bal_ex_mort  total_bc_limit  \
count     2.190392e+06       2.210638e+06    2.210638e+06
mean      1.782428e+05       5.102294e+04    2.319377e+04
std       1.815748e+05       4.991124e+04    2.300656e+04
min       0.000000e+00       0.000000e+00    0.000000e+00
25%       5.073100e+04       2.089200e+04    8.300000e+03
50%       1.142985e+05       3.786400e+04    1.630000e+04
75%       2.577550e+05       6.435000e+04    3.030000e+04
```

```
max       9.999999e+06       3.408095e+06    1.569000e+06

       total_il_high_credit_limit  revol_bal_joint  sec_app_inq_last_6mths  \
count                2.190392e+06     1.080200e+05           108021.000000
mean                 4.373201e+04     3.361728e+04                0.633256
std                  4.507298e+04     2.815387e+04                0.993401
min                  0.000000e+00     0.000000e+00                0.000000
25%                  1.500000e+04     1.510675e+04                0.000000
50%                  3.269600e+04     2.651650e+04                0.000000
75%                  5.880425e+04     4.376900e+04                1.000000
max                  2.118996e+06     1.110019e+06                6.000000

       sec_app_mort_acc  sec_app_open_acc  sec_app_revol_util  \
count     108021.000000     108021.000000       106184.000000
mean           1.538997         11.469455           58.169101
std            1.760569          6.627271           25.548212
min            0.000000          0.000000            0.000000
25%            0.000000          7.000000           39.800000
50%            1.000000         10.000000           60.200000
75%            2.000000         15.000000           78.600000
max           27.000000         82.000000          434.300000

       sec_app_open_act_il  sec_app_num_rev_accts  \
count        108021.000000          108021.000000
mean              3.010554              12.533072
std               3.275893               8.150964
min               0.000000               0.000000
25%               1.000000               7.000000
50%               2.000000              11.000000
75%               4.000000              17.000000
max              43.000000             106.000000

       sec_app_chargeoff_within_12_mths  sec_app_collections_12_mths_ex_med  \
count                     108021.000000                       108021.000000
mean                           0.046352                            0.077568
std                            0.411496                            0.407996
min                            0.000000                            0.000000
25%                            0.000000                            0.000000
50%                            0.000000                            0.000000
75%                            0.000000                            0.000000
max                           21.000000                           23.000000

       sec_app_mths_since_last_major_derog  deferral_term  hardship_amount  \
count                         35942.000000        10613.0     10613.000000
mean                             36.937928            3.0       155.006696
std                              23.924584            0.0       129.113137
min                               0.000000            3.0         0.640000
```

```
25%                              16.000000            3.0        59.370000
50%                              36.000000            3.0       119.040000
75%                              56.000000            3.0       213.260000
max                             185.000000            3.0       943.940000

       hardship_length  hardship_dpd  \
count          10613.0  10613.000000
mean               3.0     13.686422
std                0.0      9.728138
min                3.0      0.000000
25%                3.0      5.000000
50%                3.0     15.000000
75%                3.0     22.000000
max                3.0     37.000000

       orig_projected_additional_accrued_interest  \
count                                 8426.000000
mean                                   454.840802
std                                    375.830737
min                                      1.920000
25%                                    174.967500
50%                                    352.605000
75%                                    622.792500
max                                   2680.890000

       hardship_payoff_balance_amount  hardship_last_payment_amount  \
count                    10613.000000                  10613.000000
mean                     11628.036442                    193.606331
std                       7615.161123                    198.694368
min                         55.730000                      0.010000
25%                       5628.730000                     43.780000
50%                      10044.220000                    132.890000
75%                      16114.940000                    284.180000
max                      40306.410000                   1407.860000

       settlement_amount  settlement_percentage  settlement_term
count       33056.000000           33056.000000     33056.000000
mean         5030.606922              47.775600        13.148596
std          3692.027842               7.336379         8.192319
min            44.210000               0.200000         0.000000
25%          2227.000000              45.000000         6.000000
50%          4172.855000              45.000000        14.000000
75%          6870.782500              50.000000        18.000000
max         33601.000000             521.350000       181.000000
```

[8]:
```
df.isnull().sum()
```

[8]:

id	2260668
member_id	2260668
loan_amnt	0
funded_amnt	0
funded_amnt_inv	0
term	0
int_rate	0
installment	0
grade	0
sub_grade	0
emp_title	166969
emp_length	146907
home_ownership	0
annual_inc	4
verification_status	0
issue_d	0
loan_status	0
pymnt_plan	0
url	2260668
desc	2134603
purpose	0
title	23326
zip_code	1
addr_state	0
dti	1711
delinq_2yrs	29
earliest_cr_line	29
inq_last_6mths	30
mths_since_last_delinq	1158502
mths_since_last_record	1901512
open_acc	29
pub_rec	29
revol_bal	0
revol_util	1802
total_acc	29
initial_list_status	0
out_prncp	0
out_prncp_inv	0
total_pymnt	0
total_pymnt_inv	0
total_rec_prncp	0
total_rec_int	0
total_rec_late_fee	0
recoveries	0
collection_recovery_fee	0
last_pymnt_d	2426
last_pymnt_amnt	0

next_pymnt_d	1303607
last_credit_pull_d	73
collections_12_mths_ex_med	145
mths_since_last_major_derog	1679893
policy_code	0
application_type	0
annual_inc_joint	2139958
dti_joint	2139962
verification_status_joint	2144938
acc_now_delinq	29
tot_coll_amt	70276
tot_cur_bal	70276
open_acc_6m	866130
open_act_il	866129
open_il_12m	866129
open_il_24m	866129
mths_since_rcnt_il	909924
total_bal_il	866129
il_util	1068850
open_rv_12m	866129
open_rv_24m	866129
max_bal_bc	866129
all_util	866348
total_rev_hi_lim	70276
inq_fi	866129
total_cu_tl	866130
inq_last_12m	866130
acc_open_past_24mths	50030
avg_cur_bal	70346
bc_open_to_buy	74935
bc_util	76071
chargeoff_within_12_mths	145
delinq_amnt	29
mo_sin_old_il_acct	139071
mo_sin_old_rev_tl_op	70277
mo_sin_rcnt_rev_tl_op	70277
mo_sin_rcnt_tl	70276
mort_acc	50030
mths_since_recent_bc	73412
mths_since_recent_bc_dlq	1740967
mths_since_recent_inq	295435
mths_since_recent_revol_delinq	1520309
num_accts_ever_120_pd	70276
num_actv_bc_tl	70276
num_actv_rev_tl	70276
num_bc_sats	58590
num_bc_tl	70276

num_il_tl	70276
num_op_rev_tl	70276
num_rev_accts	70277
num_rev_tl_bal_gt_0	70276
num_sats	58590
num_tl_120dpd_2m	153657
num_tl_30dpd	70276
num_tl_90g_dpd_24m	70276
num_tl_op_past_12m	70276
pct_tl_nvr_dlq	70431
percent_bc_gt_75	75379
pub_rec_bankruptcies	1365
tax_liens	105
tot_hi_cred_lim	70276
total_bal_ex_mort	50030
total_bc_limit	50030
total_il_high_credit_limit	70276
revol_bal_joint	2152648
sec_app_earliest_cr_line	2152647
sec_app_inq_last_6mths	2152647
sec_app_mort_acc	2152647
sec_app_open_acc	2152647
sec_app_revol_util	2154484
sec_app_open_act_il	2152647
sec_app_num_rev_accts	2152647
sec_app_chargeoff_within_12_mths	2152647
sec_app_collections_12_mths_ex_med	2152647
sec_app_mths_since_last_major_derog	2224726
hardship_flag	0
hardship_type	2250055
hardship_reason	2250055
hardship_status	2250055
deferral_term	2250055
hardship_amount	2250055
hardship_start_date	2250055
hardship_end_date	2250055
payment_plan_start_date	2250055
hardship_length	2250055
hardship_dpd	2250055
hardship_loan_status	2250055
orig_projected_additional_accrued_interest	2252242
hardship_payoff_balance_amount	2250055
hardship_last_payment_amount	2250055
disbursement_method	0
debt_settlement_flag	0
debt_settlement_flag_date	2227612
settlement_status	2227612

```
settlement_date                                 2227612
settlement_amount                               2227612
settlement_percentage                           2227612
settlement_term                                 2227612
dtype: int64
```

0.0.2 2. Initial Data Quality Assessment

[9]:
```
missing_value_summary=pd.DataFrame({'missing_count':df.isnull().
  ↪sum(),'missing_pct':df.isnull().mean()*100})
missing_value_summary=missing_value_summary.sort_values(by='missing_pct',␣
  ↪ascending=False).reset_index()
display(missing_value_summary)
```

	index	missing_count	missing_pct
0	id	2260668	100.000000
1	url	2260668	100.000000
2	member_id	2260668	100.000000
3	orig_projected_additional_accrued_interest	2252242	99.627278
4	hardship_length	2250055	99.530537
5	hardship_reason	2250055	99.530537
6	hardship_status	2250055	99.530537
7	deferral_term	2250055	99.530537
8	hardship_amount	2250055	99.530537
9	hardship_start_date	2250055	99.530537
10	hardship_end_date	2250055	99.530537
11	payment_plan_start_date	2250055	99.530537
12	hardship_dpd	2250055	99.530537
13	hardship_loan_status	2250055	99.530537
14	hardship_payoff_balance_amount	2250055	99.530537
15	hardship_last_payment_amount	2250055	99.530537
16	hardship_type	2250055	99.530537
17	debt_settlement_flag_date	2227612	98.537777
18	settlement_status	2227612	98.537777
19	settlement_date	2227612	98.537777
20	settlement_amount	2227612	98.537777
21	settlement_percentage	2227612	98.537777
22	settlement_term	2227612	98.537777
23	sec_app_mths_since_last_major_derog	2224726	98.410116
24	sec_app_revol_util	2154484	95.302981
25	revol_bal_joint	2152648	95.221766
26	sec_app_open_acc	2152647	95.221722
27	sec_app_earliest_cr_line	2152647	95.221722
28	sec_app_collections_12_mths_ex_med	2152647	95.221722
29	sec_app_chargeoff_within_12_mths	2152647	95.221722
30	sec_app_num_rev_accts	2152647	95.221722
31	sec_app_open_act_il	2152647	95.221722

32	sec_app_mort_acc	2152647	95.221722
33	sec_app_inq_last_6mths	2152647	95.221722
34	verification_status_joint	2144938	94.880717
35	dti_joint	2139962	94.660605
36	annual_inc_joint	2139958	94.660428
37	desc	2134603	94.423551
38	mths_since_last_record	1901512	84.112837
39	mths_since_recent_bc_dlq	1740967	77.011175
40	mths_since_last_major_derog	1679893	74.309585
41	mths_since_recent_revol_delinq	1520309	67.250432
42	next_pymnt_d	1303607	57.664681
43	mths_since_last_delinq	1158502	51.246003
44	il_util	1068850	47.280273
45	mths_since_rcnt_il	909924	40.250227
46	all_util	866348	38.322655
47	open_acc_6m	866130	38.313012
48	inq_last_12m	866130	38.313012
49	total_cu_tl	866130	38.313012
50	total_bal_il	866129	38.312968
51	open_rv_12m	866129	38.312968
52	inq_fi	866129	38.312968
53	max_bal_bc	866129	38.312968
54	open_rv_24m	866129	38.312968
55	open_act_il	866129	38.312968
56	open_il_12m	866129	38.312968
57	open_il_24m	866129	38.312968
58	mths_since_recent_inq	295435	13.068482
59	emp_title	166969	7.385826
60	num_tl_120dpd_2m	153657	6.796973
61	emp_length	146907	6.498389
62	mo_sin_old_il_acct	139071	6.151766
63	bc_util	76071	3.364979
64	percent_bc_gt_75	75379	3.334368
65	bc_open_to_buy	74935	3.314728
66	mths_since_recent_bc	73412	3.247359
67	pct_tl_nvr_dlq	70431	3.115495
68	avg_cur_bal	70346	3.111735
69	mo_sin_old_rev_tl_op	70277	3.108683
70	mo_sin_rcnt_rev_tl_op	70277	3.108683
71	num_rev_accts	70277	3.108683
72	num_bc_tl	70276	3.108639
73	num_actv_bc_tl	70276	3.108639
74	num_accts_ever_120_pd	70276	3.108639
75	mo_sin_rcnt_tl	70276	3.108639
76	num_il_tl	70276	3.108639
77	num_op_rev_tl	70276	3.108639
78	tot_coll_amt	70276	3.108639
79	num_rev_tl_bal_gt_0	70276	3.108639

80	total_rev_hi_lim	70276	3.108639
81	tot_cur_bal	70276	3.108639
82	num_tl_30dpd	70276	3.108639
83	num_tl_90g_dpd_24m	70276	3.108639
84	num_tl_op_past_12m	70276	3.108639
85	tot_hi_cred_lim	70276	3.108639
86	total_il_high_credit_limit	70276	3.108639
87	num_actv_rev_tl	70276	3.108639
88	num_bc_sats	58590	2.591712
89	num_sats	58590	2.591712
90	total_bal_ex_mort	50030	2.213063
91	total_bc_limit	50030	2.213063
92	mort_acc	50030	2.213063
93	acc_open_past_24mths	50030	2.213063
94	title	23326	1.031819
95	last_pymnt_d	2426	0.107313
96	revol_util	1802	0.079711
97	dti	1711	0.075686
98	pub_rec_bankruptcies	1365	0.060380
99	collections_12_mths_ex_med	145	0.006414
100	chargeoff_within_12_mths	145	0.006414
101	tax_liens	105	0.004645
102	last_credit_pull_d	73	0.003229
103	inq_last_6mths	30	0.001327
104	open_acc	29	0.001283
105	total_acc	29	0.001283
106	delinq_amnt	29	0.001283
107	pub_rec	29	0.001283
108	acc_now_delinq	29	0.001283
109	earliest_cr_line	29	0.001283
110	delinq_2yrs	29	0.001283
111	annual_inc	4	0.000177
112	zip_code	1	0.000044
113	verification_status	0	0.000000
114	home_ownership	0	0.000000
115	disbursement_method	0	0.000000
116	sub_grade	0	0.000000
117	grade	0	0.000000
118	installment	0	0.000000
119	loan_status	0	0.000000
120	debt_settlement_flag	0	0.000000
121	int_rate	0	0.000000
122	term	0	0.000000
123	funded_amnt_inv	0	0.000000
124	funded_amnt	0	0.000000
125	loan_amnt	0	0.000000
126	issue_d	0	0.000000
127	revol_bal	0	0.000000

128	pymnt_plan	0	0.000000
129	total_rec_int	0	0.000000
130	application_type	0	0.000000
131	policy_code	0	0.000000
132	last_pymnt_amnt	0	0.000000
133	collection_recovery_fee	0	0.000000
134	recoveries	0	0.000000
135	total_rec_late_fee	0	0.000000
136	total_rec_prncp	0	0.000000
137	purpose	0	0.000000
138	total_pymnt_inv	0	0.000000
139	total_pymnt	0	0.000000
140	out_prncp_inv	0	0.000000
141	out_prncp	0	0.000000
142	initial_list_status	0	0.000000
143	hardship_flag	0	0.000000
144	addr_state	0	0.000000

[10]:
```
# Check duplicate rows
print("Duplicate rows:", df.duplicated().sum())
```

```
Duplicate rows: 0
```

[11]:
```
# Identify columns containing only one unique value
constant_cols = df.columns[df.nunique(dropna=False) == 1].tolist()
print("Constant columns:", constant_cols)
```

```
Constant columns: ['id', 'member_id', 'url', 'policy_code']
```

0.0.3 3. Data Cleaning and Preparation

[12]:
```
# Remove constant columns as well as irrelevant columns
cols_to_drop = ['id', 'member_id', 'url', 'policy_code', 'desc']

df.drop(columns=cols_to_drop, inplace=True, errors='ignore')
```

[13]:
```
# Convett Date columns
date_cols = ['issue_d', 'earliest_cr_line','last_pymnt_d',␣
 ↪'next_pymnt_d','last_credit_pull_d','hardship_start_date',␣
 ↪'hardship_end_date','payment_plan_start_date','debt_settlement_flag_date',␣
 ↪'settlement_date']

for col in date_cols:
    if col in df.columns:
        if not pd.api.types.is_datetime64_any_dtype(df[col]):
            df[col] = pd.to_datetime(df[col],format='%b-%Y', errors='coerce')
```

[14]:
```
# Convert term to integer
df['term'] = pd.to_numeric(df['term'].astype('string').str.strip().str.
 ↪replace(' months', '', regex=False), errors='coerce').astype('Int64')
```

Handle missing values by category

Instead of 75 individual checks, we'll use one consistent policy.

Feature category Action Reason

- Joint application fields Keep NaN Not applicable to individual loans
- Hardship and settlement fields Keep NaN Not applicable to most loans
- Historical credit metrics (~38% missing) Keep NaN Largely unavailable before 2016
- Months-since-delinquency fields Keep NaN Absence may reflect no recorded event
- emp_length, emp_title Label missing as Unknown Categorical information
- annual_inc, dti, other numeric fields Preserve missing initially Avoid introducing artificial values
- next_pymnt_d Keep NaT No scheduled payment date may be meaningful

[15]:
```
for col in ['grade', 'home_ownership', 'verification_status',␣
 ↪'application_type']:
    print(col, df[col].value_counts(dropna=False))
```

```
grade grade
B    663557
C    650053
A    433027
D    324424
E    135639
F     41800
G     12168
Name: count, dtype: int64
home_ownership home_ownership
MORTGAGE    1111450
RENT         894929
OWN          253057
ANY             996
OTHER           182
NONE             54
Name: count, dtype: int64
verification_status verification_status
Source Verified    886231
Not Verified       744806
Verified           629631
Name: count, dtype: int64
application_type application_type
Individual    2139958
```

```
Joint App      120710
Name: count, dtype: int64
```

[16]:
```
# Fill missing employment categories
categorical_fill = ['emp_length', 'emp_title']

for col in categorical_fill:
    df[col] = df[col].fillna('Unknown')

# Observation: I am not filling every numerical missing value because EDA can␣
 ↪handle missing observations, and arbitrary median imputation may distort the␣
 ↪loan distributions which we're trying to understand.
```

[17]:
```
df[['annual_inc', 'dti']].quantile([0.25, 0.5, 0.75, 0.95, 0.99, 0.999])
```

[17]:

	annual_inc	dti
0.250	46000.0	11.89000
0.500	65000.0	17.84000
0.750	93000.0	24.49000
0.950	160000.0	33.88000
0.990	270000.0	42.71000
0.999	600000.0	106.63132

[18]:
```
df.loc[df['dti'] < 0, 'dti'] = np.nan
```

[19]:
```
income_limit = df['annual_inc'].quantile(0.999)

print("Income threshold:", income_limit)
print("Records above threshold:", (df['annual_inc'] > income_limit).sum())
```

```
Income threshold: 600000.0
Records above threshold: 2165
```

[20]:
```
fig = px.histogram(df[df['annual_inc'] <= income_limit],x='annual_inc',␣
 ↪nbins=10,title='Annual Income Distribution')
fig.show()
```

- Annual income is right-skewed, with a median of $65,000.
- Only 2,165 records have annual income above $600,000, approximately 0.096% of the dataset.
- DTI also has extreme observations. Its 99.9th percentile is 106.63.
- These values are statistical outliers, not necessarily data errors.

[21]:
```
# winsorization (upper-tail capping) rather than row removal. It retains all␣
 ↪observations while limiting the influence of extreme values.

df['annual_inc_capped'] = df['annual_inc'].clip(upper=600000)

df['dti_capped'] = df['dti'].clip(upper=106.63)
```

[22]:
```
df[['annual_inc_capped', 'dti_capped']].describe()
```

[22]:

	annual_inc_capped	dti_capped
count	2.260664e+06	2.258955e+06
mean	7.727262e+04	1.866940e+01
std	5.150243e+04	9.748886e+00
min	0.000000e+00	0.000000e+00
25%	4.600000e+04	1.189000e+01
50%	6.500000e+04	1.784000e+01
75%	9.300000e+04	2.449000e+01
max	6.000000e+05	1.066300e+02

4. Post-Cleaning Data Validation

[23]:
```
# Columns where negative amounts should not occur
nonnegative_cols =␣
 ↪['loan_amnt','funded_amnt','installment','annual_inc','recoveries','collection_recovery_fee

quality_checks = pd.DataFrame({'negative_values':[(df[col] < 0).sum() for col␣
 ↪in nonnegative_cols],'missing_values':[df[col].isna().sum() for col in␣
 ↪nonnegative_cols]},index=nonnegative_cols)

display(quality_checks)
```

	negative_values	missing_values
loan_amnt	0	0
funded_amnt	0	0
installment	0	0
annual_inc	0	4
recoveries	0	0
collection_recovery_fee	0	0
total_pymnt	0	0
total_rec_prncp	0	0
out_prncp	0	0

[24]:
```
print("Loan term:")
print(df['term'].value_counts(dropna=False))

print("\nLoan status:")
print(df['loan_status'].value_counts(dropna=False))

print("\nLoan grades:")
print(df['grade'].value_counts(dropna=False))
```

```
Loan term:
term
36    1609754
60     650914
Name: count, dtype: Int64
```

```
Loan status:
loan_status
Fully Paid                                                   1041952
Current                                                       919695
Charged Off                                                   261655
Late (31-120 days)                                             21897
In Grace Period                                                 8952
Late (16-30 days)                                               3737
Does not meet the credit policy. Status:Fully Paid              1988
Does not meet the credit policy. Status:Charged Off              761
Default                                                           31
Name: count, dtype: int64

Loan grades:
grade
B    663557
C    650053
A    433027
D    324424
E    135639
F     41800
G     12168
Name: count, dtype: int64
```

[25]:
```
print("Final dataset shape:", df.shape)

print("Duplicate rows:", df.duplicated().sum())

print("Total missing cells:", df.isna().sum().sum())

print("Memory usage (MB):",round(df.memory_usage(deep=True).sum() / 1024**2, 2))

display(df.dtypes.value_counts())

# Observation: Since, 99.4 million values are missing, so we cant remove them␣
 ↪as many belong to joint application, hardship, settlement, and historically␣
 ↪unavailable credit fields.
```

```
Final dataset shape: (2260668, 142)
Duplicate rows: 0
Total missing cells: 99438194
Memory usage (MB): 4560.96

float64           104
object             24
datetime64[ns]     10
int64               3
Int64               1
```

```
Name: count, dtype: int64
```

[26]:
```
print("Funded amount exceeds loan amount:", (df['funded_amnt'] >␣
↪df['loan_amnt']).sum())

print("Invalid interest rates:", ((df['int_rate'] < 0) | (df['int_rate'] >␣
↪100)).sum())

print("Principal Received > Funded:", (df['total_rec_prncp'] >␣
↪df['funded_amnt'] + 0.01).sum())
```

```
Funded amount exceeds loan amount: 0
Invalid interest rates: 0
Principal Received > Funded: 21
```

[27]:
```
principal_issue = df[df['total_rec_prncp'] > df['funded_amnt'] + 0.01].copy()

principal_issue['difference'] = principal_issue['total_rec_prncp'] -␣
↪principal_issue['funded_amnt']

display(principal_issue[['loan_status', 'funded_amnt', 'total_rec_prncp',␣
↪'difference', 'recoveries']])
```

```
                                                loan_status  funded_amnt  \
2113041                                          Fully Paid        10000
2113063                                          Fully Paid        13225
2114124                                          Fully Paid        21000
2114149                                          Fully Paid        21000
2114182                                          Fully Paid        13000
2114322                                          Fully Paid        15000
2114385                                          Fully Paid        25000
2114515                                          Fully Paid         7600
2114730                                          Fully Paid        14000
2117468                                          Fully Paid        35000
2117786                                          Fully Paid        25000
2118694                                          Fully Paid        25000
2118719                                          Fully Paid        35000
2120719                                          Fully Paid        12000
2121205                                          Fully Paid        14400
2123629                                          Fully Paid        14050
2123970                                          Fully Paid        15000
2125186                                          Fully Paid         7800
2125385                                          Fully Paid        20000
2125535                                          Fully Paid        20000
2139689  Does not meet the credit policy. Status:Fully …           24000

         total_rec_prncp  difference  recoveries
2113041         10000.03        0.03         0.0
```

```
2113063         13225.03        0.03          0.0
2114124         21000.03        0.03          0.0
2114149         21000.02        0.02          0.0
2114182         13000.04        0.04          0.0
2114322         15000.05        0.05          0.0
2114385         25000.05        0.05          0.0
2114515          7600.08        0.08          0.0
2114730         14000.07        0.07          0.0
2117468         35000.02        0.02          0.0
2117786         25000.03        0.03          0.0
2118694         25000.02        0.02          0.0
2118719         35000.02        0.02          0.0
2120719         12000.02        0.02          0.0
2121205         14400.02        0.02          0.0
2123629         14050.11        0.11          0.0
2123970         15000.02        0.02          0.0
2125186          7800.25        0.25          0.0
2125385         20000.24        0.24          0.0
2125535         20000.24        0.24          0.0
2139689         24000.06        0.06          0.0
```

[28]:
```
principal_issue['difference'].describe()
```

[28]:
```
count    21.000000
mean      0.069048
std       0.076675
min       0.020000
25%       0.020000
50%       0.030000
75%       0.070000
max       0.250000
Name: difference, dtype: float64
```

Financial Data Quality Validation Identified 21 loans where recorded principal repayments exceeded funded amounts by \$0.02–\$0.25. All affected loans were classified as fully paid and had zero recorded recoveries. The discrepancies were retained and documented as minor financial reconciliation exceptions, potentially attributable to rounding or payment allocation differences. No records were removed.

Saving the cleaned DataFrame because this dataset contains 2.26 million rows and 140 columns, occupying approximately 4.5 GB of RAM and Without saving the prepared dataset, every time after restarting Jupyter, we would need to repeat entire data cleaning operations. But, with Pickle, we perform these steps once and save the result in my directory.

[29]:
```
df.to_pickle("lendingclub_prepared.pkl")
```

5. Exploratory Data Analysis (EDA)

1. Loan Status Distribution

[30]:
```
# Number of loans in each category
loan_status_df = df['loan_status'].value_counts().reset_index()

# Percentage of loans in each category
loan_status_df['percentage'] = (loan_status_df['count'] /␣
 ↪loan_status_df['count'].sum() * 100).round(2)
loan_status_df
```

[30]:

	loan_status	count	percentage
0	Fully Paid	1041952	46.09
1	Current	919695	40.68
2	Charged Off	261655	11.57
3	Late (31-120 days)	21897	0.97
4	In Grace Period	8952	0.40
5	Late (16-30 days)	3737	0.17
6	Does not meet the credit policy. Status:Fully …	1988	0.09
7	Does not meet the credit policy. Status:Charge…	761	0.03
8	Default	31	0.00

[31]:
```
fig1 = px.bar(loan_status_df, x='count', y='loan_status',␣
 ↪orientation='h',hover_data=['percentage'], title='Distribution of Loan␣
 ↪Status')

fig1.update_layout(yaxis={'categoryorder': 'total ascending'})
fig1.show()
```

Observation 1: Loan Status Distribution

The portfolio is dominated by Fully Paid (46.09%) and Current (40.68%) loans. Charged Off loans account for approximately 11.6% of all records, representing a substantial segment for further recovery analysis. Since many loans remain active, the observed charge-off proportion should not be interpreted as the final default rate.

2. Loan Amount Distribution

[32]:
```
pd.options.display.float_format = '{:,.2f}'.format
```

[33]:
```
fig2 = px.histogram(df, x='loan_amnt', nbins=10, title='Distribution of Loan␣
 ↪Amounts')
fig2.show()

df['loan_amnt'].describe()
```

[33]:
```
count   2,260,668.00
mean       15,046.93
std         9,190.25
min           500.00
25%         8,000.00
50%        12,900.00
```

```
75%         20,000.00
max         40,000.00
Name: loan_amnt, dtype: float64
```

Observation 2: Loan Amount Distribution

The average loan amount is approximately 15,047, while the median is 12,900, suggesting a right-skewed distribution. Half of the loans range between 8,000 and 20,000, indicating that loan amounts are concentrated in the lower to middle portion of the observed range.

[ ]:

3. Loan Amount by Loan Status

[34]:
```
fig3 = px.box(df, x='loan_amnt', y='loan_status', title='Loan Amount by Loan␣
↪Status')
fig3.show()
```

Observation 3: Loan Amount by Loan Status * Charged Off and Fully Paid: Their loan amount distributions appear broadly similar, with median loan amounts around $13K–$15K. * Late loans: Loans classified as Late (31–120 days) appear to have slightly higher median loan amounts. * Variation: Most major loan-status categories include loans ranging from approximately 500 to 40,000. * Credit policy exceptions: These two categories appear to have lower loan amounts than most other groups.

4. Interest Rate Distribution

[35]:
```
fig4 = px.histogram(df, x='int_rate', nbins=10, title='Distribution of Interest␣
↪Rates')
fig4.show()

df['int_rate'].describe()
```

[35]:
```
count   2,260,668.00
mean           13.09
std             4.83
min             5.31
25%             9.49
50%            12.62
75%            15.99
max            30.99
Name: int_rate, dtype: float64
```

Observation 4:

- Highest concentration: Most loans have interest rates between approximately 5% and 20%.
- Most frequent range: The 10%–15% interval contains the largest number of loans.
- Right-skewed distribution: The number of loans generally decreases as interest rates increase beyond 15%.
- High-interest loans: Relatively few loans have interest rates above 25%.

Business interpretation: This portfolio contains substantially more low-to-moderate-interest loans than high-interest loans. However, we cannot determine credit risk or recovery performance from interest rates alone.

5. Interest Rate by Loan Grade

[36]:
```
fig5 = px.box(df, x='int_rate', y='grade',
              category_orders={'grade': list('ABCDEFG')},
              title='Interest Rate by Loan Grade')
fig5.show()
```

Observation 5:

- Grade A: Lowest interest rates, with a median around 7%.
- Grades B and C: Median interest rates increase to approximately 11% and 14%.
- Grades D and E: Interest rates increase further, with medians around 18% and 21%.
- Grades F and G: Highest interest rates, with Grade G approaching 28%.

Business interpretation: Interest rates increase consistently from Grade A to Grade G, reflecting LendingClub's risk-based loan pricing. Higher-risk grades generally carry higher interest rates.

6. Loan Grade vs. Charge-Off Rate

[37]:
```
df['is_charged_off'] = df['loan_status'].str.contains('Charged Off', na=False)

chargeoff_by_grade = (df.groupby('grade')['is_charged_off'].mean() * 100).
 ↪round(2)

fig = px.bar(chargeoff_by_grade.reset_index(), x='grade', y='is_charged_off',␣
 ↪title='Observed Charge-Off Percentage by Loan Grade')
fig.show()
```

Observation 6:

- Grade A has the lowest observed charge-off percentage.
- Grades F and G have the highest, with more than one-third of loans classified as charged off.
- This supports the relationship between LendingClub's risk grades and observed loan performance.

0.0.4 Recovery on Charged-Off Loans

[38]:
```
# Creating a new DataFrame for Chargedoff Loans

charged_off_df = df[df['is_charged_off']].copy()


print("Charged-off loans:", len(charged_off_df))

charged_off_df.head()
```

```
Charged-off loans: 262416
```

[38]:

	loan_amnt	funded_amnt	funded_amnt_inv	term	int_rate	installment
5040	8000	8000	8,000.00	36	6.46	245.05
25963	6000	6000	6,000.00	36	14.47	206.44
41393	10000	10000	10,000.00	36	8.81	317.12
55148	10000	10000	10,000.00	60	27.27	306.97
58105	35000	35000	35,000.00	36	16.14	1,232.92

	grade	sub_grade	emp_title	emp_length	home_ownership
5040	A	A1	Meat Cutter	10+ years	MORTGAGE
25963	C	C2	Unknown	Unknown	MORTGAGE
41393	A	A5	Unknown	Unknown	MORTGAGE
55148	E	E5	CNC Robotic Welder	10+ years	RENT
58105	C	C4	Unknown	< 1 year	MORTGAGE

	annual_inc	verification_status	issue_d	loan_status	pymnt_plan
5040	55,000.00	Not Verified	2018-12-01	Charged Off	n
25963	17,000.00	Not Verified	2018-12-01	Charged Off	n
41393	56,000.00	Verified	2018-11-01	Charged Off	n
55148	70,000.00	Source Verified	2018-11-01	Charged Off	n
58105	300,000.00	Source Verified	2018-11-01	Charged Off	n

	purpose	title	zip_code	addr_state	dti
5040	credit_card	Credit card refinancing	984xx	WA	17.83
25963	debt_consolidation	Debt consolidation	342xx	FL	19.99
41393	debt_consolidation	Debt consolidation	852xx	AZ	20.53
55148	debt_consolidation	Debt consolidation	612xx	IL	11.37
58105	other	Other	989xx	WA	13.88

	delinq_2yrs	earliest_cr_line	inq_last_6mths	mths_since_last_delinq
5040	0.00	2002-05-01	0.00	55.00
25963	1.00	2000-11-01	0.00	23.00
41393	0.00	2011-04-01	1.00	NaN
55148	0.00	2008-11-01	1.00	60.00
58105	0.00	2010-08-01	2.00	62.00

	mths_since_last_record	open_acc	pub_rec	revol_bal	revol_util
5040	NaN	12.00	0.00	10999	29.10
25963	NaN	8.00	0.00	650	8.30
41393	93.00	7.00	1.00	10494	53.30
55148	NaN	4.00	0.00	3193	84.00
58105	100.00	22.00	1.00	44283	84.50

	total_acc	initial_list_status	out_prncp	out_prncp_inv	total_pymnt
5040	49.00	w	0.00	0.00	0.00
25963	10.00	f	0.00	0.00	0.00
41393	19.00	w	0.00	0.00	0.00
55148	24.00	w	0.00	0.00	284.25

```
58105     44.00                   w       0.00           0.00         0.00

       total_pymnt_inv  total_rec_prncp  total_rec_int  total_rec_late_fee  \
5040              0.00             0.00           0.00                0.00
25963             0.00             0.00           0.00                0.00
41393             0.00             0.00           0.00                0.00
55148           284.25           79.72         204.53                0.00
58105             0.00             0.00           0.00                0.00

       recoveries  collection_recovery_fee last_pymnt_d  last_pymnt_amnt  \
5040         0.00                     0.00          NaT             0.00
25963        0.00                     0.00          NaT             0.00
41393        0.00                     0.00          NaT             0.00
55148        0.00                     0.00   2018-12-01           306.97
58105        0.00                     0.00          NaT             0.00

      next_pymnt_d last_credit_pull_d  collections_12_mths_ex_med  \
5040           NaT         2019-02-01                        0.00
25963          NaT         2019-02-01                        1.00
41393          NaT         2019-01-01                        0.00
55148          NaT         2019-02-01                        0.00
58105          NaT         2019-02-01                        0.00

       mths_since_last_major_derog application_type  annual_inc_joint  \
5040                           NaN       Individual               NaN
25963                        23.00       Individual               NaN
41393                          NaN       Individual               NaN
55148                        60.00       Individual               NaN
58105                        62.00       Individual               NaN

       dti_joint verification_status_joint  acc_now_delinq  tot_coll_amt  \
5040         NaN                      NaN            0.00          0.00
25963        NaN                      NaN            0.00        624.00
41393        NaN                      NaN            0.00          0.00
55148        NaN                      NaN            0.00          0.00
58105        NaN                      NaN            0.00          0.00

       tot_cur_bal  open_acc_6m  open_act_il  open_il_12m  open_il_24m  \
5040    190,171.00         1.00         2.00         1.00         2.00
25963     5,005.00         0.00         1.00         1.00         1.00
41393   301,829.00         1.00         3.00         1.00         2.00
55148    23,289.00         2.00         2.00         3.00         5.00
58105    87,111.00         0.00         4.00         1.00         2.00

       mths_since_rcnt_il  total_bal_il  il_util  open_rv_12m  open_rv_24m  \
5040                11.00      9,911.00    43.00         4.00         8.00
25963               11.00      4,355.00    81.00         1.00         2.00
```

```
41393              11.00     20,277.00    59.00         0.00         2.00
55148               1.00     20,096.00    93.00         0.00         2.00
58105              12.00     42,828.00    69.00         1.00         2.00

       max_bal_bc  all_util  total_rev_hi_lim  inq_fi  total_cu_tl  \
5040     4,663.00     34.00         37,800.00    0.00        15.00
25963      202.00     38.00          7,800.00    3.00         0.00
41393    5,295.00     57.00         19,700.00    3.00         1.00
55148    1,568.00     91.00          3,800.00    0.00        11.00
58105    4,064.00     77.00         52,400.00    1.00         0.00

       inq_last_12m  acc_open_past_24mths  avg_cur_bal  bc_open_to_buy  \
5040           2.00                 12.00    15,848.00        7,592.00
25963          1.00                  3.00       626.00        3,759.00
41393          3.00                  5.00    43,118.00        2,323.00
55148          2.00                  7.00     5,822.00          232.00
58105          5.00                  4.00     3,960.00        5,173.00

       bc_util  chargeoff_within_12_mths  delinq_amnt  mo_sin_old_il_acct  \
5040     54.50                     0.00         0.00              151.00
25963     8.30                     0.00         0.00               11.00
41393    80.60                     0.00         0.00               91.00
55148    87.10                     0.00         0.00              120.00
58105    74.40                     0.00         0.00               92.00

       mo_sin_old_rev_tl_op  mo_sin_rcnt_rev_tl_op  mo_sin_rcnt_tl  mort_acc  \
5040                 161.00                   3.00            3.00      5.00
25963                140.00                  11.00           11.00      0.00
41393                 71.00                  19.00            6.00      3.00
55148                 89.00                  17.00            1.00      0.00
58105                 99.00                  12.00           12.00      0.00

       mths_since_recent_bc  mths_since_recent_bc_dlq  mths_since_recent_inq  \
5040                   7.00                       NaN                  11.00
25963                 21.00                       NaN                  10.00
41393                 19.00                       NaN                   5.00
55148                 17.00                     60.00                   3.00
58105                 26.00                       NaN                   1.00

       mths_since_recent_revol_delinq  num_accts_ever_120_pd  num_actv_bc_tl  \
5040                            55.00                   0.00            3.00
25963                             NaN                   2.00            3.00
41393                             NaN                   0.00            2.00
55148                           60.00                   2.00            1.00
58105                           78.00                   1.00            9.00

       num_actv_rev_tl  num_bc_sats  num_bc_tl  num_il_tl  num_op_rev_tl  \
```

```
5040              4.00         4.00      14.00      13.00           9.00
25963             4.00         4.00       4.00       1.00           7.00
41393             3.00         2.00       2.00      10.00           3.00
55148             2.00         1.00       3.00      20.00           2.00
58105            15.00         9.00      16.00      10.00          18.00

       num_rev_accts  num_rev_tl_bal_gt_0  num_sats  num_tl_120dpd_2m  \
5040           31.00                 4.00     12.00              0.00
25963           7.00                 4.00      8.00              0.00
41393           6.00                 3.00      7.00              0.00
55148           4.00                 2.00      4.00              0.00
58105          34.00                15.00     22.00              0.00

       num_tl_30dpd  num_tl_90g_dpd_24m  num_tl_op_past_12m  pct_tl_nvr_dlq  \
5040           0.00                0.00                6.00           98.00
25963          0.00                1.00                2.00           80.00
41393          0.00                0.00                2.00          100.00
55148          0.00                0.00                3.00           91.70
58105          0.00                0.00                2.00           95.50

       percent_bc_gt_75  pub_rec_bankruptcies  tax_liens  tot_hi_cred_lim  \
5040              25.00                  0.00       0.00       232,617.00
25963              0.00                  0.00       0.00        13,200.00
41393             50.00                  1.00       0.00       327,790.00
55148            100.00                  0.00       0.00        25,493.00
58105             44.40                  1.00       0.00       142,518.00

       total_bal_ex_mort  total_bc_limit  total_il_high_credit_limit  \
5040           20,910.00       16,700.00                   23,181.00
25963           5,005.00        4,100.00                    5,400.00
41393          30,771.00       12,000.00                   34,590.00
55148          23,289.00        1,800.00                   21,693.00
58105          87,111.00       20,200.00                   90,118.00

       revol_bal_joint sec_app_earliest_cr_line  sec_app_inq_last_6mths  \
5040               NaN                      NaN                     NaN
25963              NaN                      NaN                     NaN
41393              NaN                      NaN                     NaN
55148              NaN                      NaN                     NaN
58105              NaN                      NaN                     NaN

       sec_app_mort_acc  sec_app_open_acc  sec_app_revol_util  \
5040                NaN               NaN                 NaN
25963               NaN               NaN                 NaN
41393               NaN               NaN                 NaN
55148               NaN               NaN                 NaN
58105               NaN               NaN                 NaN
```

```
       sec_app_open_act_il  sec_app_num_rev_accts  \
5040                   NaN                   NaN
25963                  NaN                   NaN
41393                  NaN                   NaN
55148                  NaN                   NaN
58105                  NaN                   NaN

       sec_app_chargeoff_within_12_mths  sec_app_collections_12_mths_ex_med  \
5040                                NaN                                 NaN
25963                               NaN                                 NaN
41393                               NaN                                 NaN
55148                               NaN                                 NaN
58105                               NaN                                 NaN

       sec_app_mths_since_last_major_derog hardship_flag hardship_type  \
5040                                   NaN             N           NaN
25963                                  NaN             N           NaN
41393                                  NaN             N           NaN
55148                                  NaN             N           NaN
58105                                  NaN             N           NaN

      hardship_reason hardship_status  deferral_term  hardship_amount  \
5040              NaN             NaN            NaN              NaN
25963             NaN             NaN            NaN              NaN
41393             NaN             NaN            NaN              NaN
55148             NaN             NaN            NaN              NaN
58105             NaN             NaN            NaN              NaN

      hardship_start_date hardship_end_date payment_plan_start_date  \
5040                  NaT               NaT                     NaT
25963                 NaT               NaT                     NaT
41393                 NaT               NaT                     NaT
55148                 NaT               NaT                     NaT
58105                 NaT               NaT                     NaT

       hardship_length  hardship_dpd hardship_loan_status  \
5040               NaN           NaN                  NaN
25963              NaN           NaN                  NaN
41393              NaN           NaN                  NaN
55148              NaN           NaN                  NaN
58105              NaN           NaN                  NaN

       orig_projected_additional_accrued_interest  \
5040                                          NaN
25963                                         NaN
41393                                         NaN
```

```
55148                                        NaN
58105                                        NaN

       hardship_payoff_balance_amount  hardship_last_payment_amount  \
5040                              NaN                           NaN
25963                             NaN                           NaN
41393                             NaN                           NaN
55148                             NaN                           NaN
58105                             NaN                           NaN

      disbursement_method debt_settlement_flag debt_settlement_flag_date  \
5040            DirectPay                    N                       NaT
25963                Cash                    N                       NaT
41393                Cash                    N                       NaT
55148                Cash                    N                       NaT
58105                Cash                    N                       NaT

      settlement_status settlement_date  settlement_amount  \
5040                NaN             NaT                NaN
25963               NaN             NaT                NaN
41393               NaN             NaT                NaN
55148               NaN             NaT                NaN
58105               NaN             NaT                NaN

       settlement_percentage  settlement_term  annual_inc_capped  dti_capped  \
5040                     NaN              NaN          55,000.00       17.83
25963                    NaN              NaN          17,000.00       19.99
41393                    NaN              NaN          56,000.00       20.53
55148                    NaN              NaN          70,000.00       11.37
58105                    NaN              NaN         300,000.00       13.88

       is_charged_off
5040             True
25963            True
41393            True
55148            True
58105            True
```

[39]:
```
charged_off_df.shape
```

[39]:
```
(262416, 143)
```

1. Recovery in Chargedoff accounts

[40]:
```
# How many charged-off loans have recovered any money.
charged_off_df['has_recovery'] = charged_off_df['recoveries'] > 0

recovery_df = charged_off_df['has_recovery'].value_counts().reset_index()
```

```
recovery_df['percentage'] = (recovery_df['count'] / recovery_df['count'].sum()␣
 ↪* 100).round(2)

fig = px.bar(recovery_df, x='has_recovery', y='count',␣
 ↪hover_data=['percentage'], title='Recovery in Charged-Off Loans')
fig.show()
```

[41]:
```
charged_off_df['has_recovery'].value_counts(normalize=True).mul(100).round(2)
```

[41]:
```
has_recovery
True    67.64
False   32.36
Name: proportion, dtype: float64
```

Observation:

- Approximately 175K–180K charged-off loans have positive recoveries.
- Approximately 85K charged-off loans have zero recorded recoveries.
- Roughly two-thirds of charged-off loans have some recovery.

Business interpretation: A significant proportion of charged-off loans still generate recoveries. Charge-off does not necessarily mean the lender recovers nothing. We still cannot say that having a positive recovery means the recovered amount is financially significant.

2. Recovery Amount Distribution

[42]:
```
charged_off_df['recoveries'].describe()
```

[42]:
```
count   262,416.00
mean      1,172.25
std       1,823.16
min           0.00
25%           0.00
50%         549.37
75%       1,675.66
max      39,859.55
Name: recoveries, dtype: float64
```

[43]:
```
fig = px.histogram(charged_off_df[charged_off_df['recoveries'] > 0],␣
 ↪x='recoveries', nbins=10, title='Distribution of Positive Recovery Amounts')
fig.show()
```

Observation:

- Strong right skew: The mean 1,172 is substantially greater than the median 549, indicating that some large recoveries pull the average upward.
- Low typical recovery: Half of all charged-off loans have recovered 549.37 or less.
- Recovery concentration: The histogram shows that most positive recoveries are concentrated below approximately 5,000, with relatively few large recoveries.

- High variability: The maximum recovery is approximately $39,860, much higher than the typical recovery.

3. Recovery Amount by Loan Grade

[44]:
```
recovery_by_grade = charged_off_df.groupby('grade')['recoveries'].agg(['count',␣
↪'mean', 'median']).round(2).reset_index()
display(recovery_by_grade)
```

	grade	count	mean	median
0	A	13774	788.51	154.98
1	B	51162	869.39	375.34
2	C	83410	1,057.93	518.14
3	D	59638	1,237.25	632.88
4	E	35522	1,557.31	965.80
5	F	14356	1,819.90	1,253.97
6	G	4554	1,933.05	1,412.43

[45]:
```
fig = px.bar(recovery_by_grade.reset_index(), x='grade', y='mean',␣
↪title='Average Recovery Amount by Loan Grade')
fig.show()
```

Observation: * Grade G has the highest average recovery 1,933, compared with Grade A 789. * Median recoveries also increase from A to G, so this pattern isn't driven only by a few unusually large recoveries. * Higher-risk grades are associated with larger absolute recovery amounts among charged-off loans.

4. Recovery as a percentage of the original loan amount

[46]:
```
charged_off_df['recovery_pct'] = (charged_off_df['recoveries'] /␣
↪charged_off_df['funded_amnt'] * 100)
```

[47]:
```
recovery_pct_by_grade = charged_off_df.groupby('grade')['recovery_pct'].
↪agg(['mean', 'median']).round(2).reset_index()
display(recovery_pct_by_grade)
```

	grade	mean	median
0	A	5.82	1.88
1	B	6.40	4.54
2	C	7.08	5.93
3	D	7.58	6.47
4	E	8.40	7.58
5	F	9.12	8.86
6	G	9.45	9.41

[48]:
```
fig = px.bar(recovery_pct_by_grade.reset_index(), x='grade', y='mean',␣
↪title='Average Recovery-to-Loan Ratio by Grade')
fig.show()
```

Observation:

- Grade G has the highest average ratio at 9.45%, compared with 5.82% for Grade A.
- Both mean and median ratios increase across grades, indicating a consistent pattern in this dataset.
- Grade A's median of 1.88% is considerably lower than its mean of 5.82%, suggesting a right-skewed distribution.
- Higher-risk grades show greater recorded recoveries relative to original funded amounts among charged-off loans.

5. Recovery After Collection Fees

[49]:
```
charged_off_df['net_recovery'] = (charged_off_df['recoveries'] -␣
↪charged_off_df['collection_recovery_fee'])
charged_off_df[['recoveries', 'collection_recovery_fee', 'net_recovery']].
↪describe()
```

[49]:

	recoveries	collection_recovery_fee	net_recovery
count	262,416.00	262,416.00	262,416.00
mean	1,172.25	194.64	977.62
std	1,823.16	325.13	1,514.15
min	0.00	0.00	0.00
25%	0.00	0.00	0.00
50%	549.37	70.96	459.82
75%	1,675.66	274.14	1,398.09
max	39,859.55	7,174.72	39,397.81

[50]:
```
net_recovery_by_grade = charged_off_df.groupby('grade')['net_recovery'].mean().
↪round(2)

fig = px.bar(net_recovery_by_grade.reset_index(), x='grade', y='net_recovery',␣
↪title='Average Net Recovery by Loan Grade')
fig.show()
```

Observation:

- Average net recovery: Each charged-off loan generates approximately 978 in recorded recoveries after reported collection fees.
- Collection costs: Average collection fees represent approximately 16.6% of average gross recoveries.
- Skewed distribution: The median net recovery 460 is considerably lower than the mean 978, indicating that larger recoveries increase the average.
- Grade differences: Grade G has the highest average net recovery in absolute dollars, while Grade A has the lowest.

Business interpretation: Higher-risk grades generate larger net recovery amounts among charged-off loans, but this does not necessarily mean collections are more efficient for those grades. Loan exposure and time available for recovery may differ.

6. Net Recovery Relative to Loan Amount

[51]:
```
charged_off_df['net_recovery_pct'] = (charged_off_df['net_recovery'] /␣
↪charged_off_df['funded_amnt'] * 100)
```

[52]:
```
net_recovery_pct_by_grade = charged_off_df.groupby('grade')['net_recovery_pct'].
↪agg(['mean', 'median']).round(2).reset_index()
display(net_recovery_pct_by_grade)
```

	grade	mean	median
0	A	4.87	1.58
1	B	5.34	3.85
2	C	5.89	4.98
3	D	6.33	5.40
4	E	7.02	6.26
5	F	7.65	7.30
6	G	7.91	7.76

[53]:
```
fig = px.bar(net_recovery_pct_by_grade.reset_index(), x='grade', y='mean',␣
↪title='Average Net Recovery-to-Loan Ratio by Grade')
fig.show()
```

Observation:

- Grade A has the lowest average net recovery-to-loan ratio at 4.87%.
- Grade G has the highest at 7.91%.
- The ratio increases consistently from Grade A to Grade G.
- Higher-risk grades show greater net recoveries relative to original funded amounts among charged-off loans.

Business interpretation: Higher-risk loans show higher net recovery-to-original-loan ratios, but this doesn't necessarily mean better collection effectiveness. Differences in outstanding balances, repayment histories, and time available for recovery may influence the results.

7. Recovery by Loan Term

[54]:
```
recovery_by_term = charged_off_df.groupby('term')['net_recovery_pct'].
↪agg(['count', 'mean', 'median']).round(2).reset_index()
display(recovery_by_term)
```

	term	count	mean	median
0	36	159798	5.70	4.40
1	60	102618	6.75	6.15

[55]:
```
fig = px.bar(recovery_by_term.reset_index(), x='term', y='mean', title='Average␣
↪Net Recovery-to-Loan Ratio by Term')
fig.update_xaxes(type='category')
fig.show()
```

Observation:

- 60-month loans: Average net recovery ratio is 6.75%, compared with 5.70% for 36-month loans.

- Difference: 1.05 percentage points.
- Median: 60-month loans also have a higher median, suggesting the difference is not driven solely by unusually large recoveries.

Business interpretation: Longer-term charged-off loans show higher net recoveries relative to their original funded amounts. However, this doesn't establish better collection efficiency because loan grades, outstanding balances, and recovery timelines may differ.

8. Recovery by Loan Issue Year

[56]:
```
charged_off_df['issue_year'] = charged_off_df['issue_d'].dt.year
```

[57]:
```
recovery_by_year = charged_off_df.groupby('issue_year')['net_recovery_pct'].
 ↪agg(['count', 'mean', 'median']).round(2).reset_index()
display(recovery_by_year)
```

	issue_year	count	mean	median
0	2007	158	4.77	1.92
1	2008	496	4.35	0.63
2	2009	723	4.56	1.45
3	2010	1757	4.72	2.55
4	2011	3297	5.10	3.60
5	2012	8644	5.89	4.93
6	2013	21022	6.98	5.89
7	2014	41056	6.91	5.62
8	2015	75275	6.47	5.26
9	2016	66667	6.17	6.05
10	2017	36379	5.01	0.00
11	2018	6942	1.64	0.00

[58]:
```
fig = px.line(recovery_by_year.reset_index(), x='issue_year', y='mean',␣
 ↪markers=True, title='Average Net Recovery-to-Loan Ratio by Issue Year')
fig.show()
```

Observation:

Loans issued in 2017 or 2018 had much less time to become charged off and accumulate recoveries than loans issued in earlier years. This is particularly noticeable in your median values: - 2016: 6.05% - 2017: 0% - 2018: 0%

A zero median means at least half of the charged-off loans in those issue-year groups had no recorded net recovery at the dataset snapshot. However, we cannot conclude that collection performance deteriorated in 2017–2018 without accounting for the time available to recover money.

9. Recovery Occurrence by Issue Year

[59]:
```
recovery_occurrence_year = (charged_off_df.
 ↪groupby('issue_year')['has_recovery'].mean().mul(100).round(2).reset_index())

fig = px.line(recovery_occurrence_year, x='issue_year', y='has_recovery',␣
 ↪markers=True, title='Recovery Occurrence (%) by Issue Year')
```

```
fig.show()
```

Observation:

Business interpretation - Nearly all charged-off loans originated between 2007 and 2012 have recorded recoveries. - Recovery occurrence decreases substantially for loans originated after 2013. - The sharp decline for 2017–2018 loans is consistent with limited time available for collections, although the chart alone cannot establish the cause.

We can not conclude that LendingClub's collection performance deteriorated. Recent loans had shorter observation periods, so recovery occurrence is not directly comparable across issue years.

10. Does Debt Settlement Affect Recovery?

[60]:
```
settlement_summary = charged_off_df.groupby('debt_settlement_flag').
 ↪agg(loans=('recoveries', 'size'),recovery_occurrence=('has_recovery',␣
 ↪'mean'),avg_net_recovery_pct=('net_recovery_pct', 'mean')).round(3).
 ↪reset_index()

settlement_summary['recovery_occurrence'] *= 100
display(settlement_summary.round(2))
```

	debt_settlement_flag	loans	recovery_occurrence	avg_net_recovery_pct
0	N	230421	65.10	5.18
1	Y	31995	86.10	12.88

[61]:
```
fig = px.bar(settlement_summary.reset_index(),x='debt_settlement_flag',␣
 ↪y='avg_net_recovery_pct', title='Average Net Recovery-to-Loan Ratio by␣
 ↪Settlement Status')
fig.show()
```

Observation:

- Charged-off loans flagged for debt settlement have an average net recovery ratio more than twice that of loans without a settlement flag.
- Debt settlement status is strongly associated with recorded net recoveries in this dataset.

However, we cannot conclude that debt settlement causes higher recovery. Settlement loans may differ in age, borrower risk, and collection history. We should also verify whether reported recoveries includes settlement payments before interpreting the financial relationship.

11. recovery occurrence by settlement status

[62]:
```
display(settlement_summary)
```

	debt_settlement_flag	loans	recovery_occurrence	avg_net_recovery_pct
0	N	230421	65.10	5.17
1	Y	31995	86.10	12.88

[63]:
```
fig = px.bar(settlement_summary.reset_index(),x='debt_settlement_flag',␣
 ↪y='recovery_occurrence', title='Recovery Occurrence by Settlement Status')
fig.show()
```

Observation:

- Approximately 86% of settlement-flagged loans have positive recoveries, compared with 65% of loans without a settlement flag.
- Settlement-flagged loans also showed higher average net recovery-to-original-loan ratios in our previous analysis.

However, these results indicate association, not causation. Settlement arrangements may be recorded after collection activity has already begun.

12. Recovery by Settlement Status

[64]:
```
charged_off_df.groupby('settlement_status', dropna=False).
↪agg(loans=('recoveries', 'size'), recovery_occurrence=('has_recovery',␣
↪'mean'), avg_net_recovery_pct=('net_recovery_pct', 'mean')).round(2).
↪reset_index()
```

[64]:

	settlement_status	loans	recovery_occurrence	avg_net_recovery_pct
0	ACTIVE	13757	0.81	6.01
1	BROKEN	4721	0.85	7.14
2	COMPLETE	13517	0.92	21.86
3	NaN	230421	0.65	5.17

Observation:

1. Completed settlements show the strongest recovery outcomes.

- 92% of completed-settlement loans have positive recorded recoveries.
- Their average net recovery-to-original-loan ratio is 21.86%, compared with 5.17% for loans without recorded settlements.
- That's approximately 4.2 times higher.

2. Active and broken settlements show much lower net recovery ratios.

- ACTIVE: 6.01%
- BROKEN: 7.14%
- COMPLETE: 21.86%

3. Settlement completion is strongly associated with recovery outcomes.

However, we cannot conclude that completing a settlement causes higher recovery. Settlement status is an outcome of the collections process, and completed settlements are likely to have more recorded payments.

Also, the 21.86% represents net recovery relative to the original funded loan amount, not the outstanding balance at charge-off.

13. Settlement Amount vs. Recovery

[65]:
```
settled_df = charged_off_df[charged_off_df['settlement_status'].notna()].copy()

settled_df[['settlement_amount', 'recoveries', 'collection_recovery_fee']].
↪describe()
```

[65]:

	settlement_amount	recoveries	collection_recovery_fee
count	31,995.00	31,995.00	31,995.00
mean	5,051.67	2,415.74	434.79
std	3,687.98	3,119.54	564.16
min	44.21	0.00	0.00
25%	2,243.10	163.27	27.49
50%	4,200.00	1,200.00	215.40
75%	6,906.00	3,524.00	633.78
max	33,601.00	27,850.00	5,632.80

[66]:
```
settled_df[['settlement_amount', 'recoveries']].corr()
```

[66]:

	settlement_amount	recoveries
settlement_amount	1.00	0.52
recoveries	0.52	1.00

Observation:

Among 31,995 charged-off loans with settlement records, settlement amounts and recorded recoveries show a moderate positive correlation (r = 0.52). The average settlement amount is 5,052, compared with average recorded recoveries of 2,416. This suggests that larger settlements are generally associated with higher recoveries, although settlement obligations and realized collections may represent different financial measures.

0.0.5 Borrower characteristics

[67]:
```
borrower_cols = ['annual_inc', 'dti', 'net_recovery_pct']

correlation = charged_off_df[borrower_cols].corr().round(2)
display(correlation)
```

	annual_inc	dti	net_recovery_pct
annual_inc	1.00	-0.13	0.03
dti	-0.13	1.00	0.01
net_recovery_pct	0.03	0.01	1.00

[68]:
```
fig = px.imshow(correlation, text_auto=True, title='Borrower Characteristics vs␣
↪Net Recovery')
fig.show()
```

[69]:
```
correlation['net_recovery_pct'].sort_values(ascending=False)
```

[69]:
```
net_recovery_pct   1.00
annual_inc         0.03
dti                0.01
Name: net_recovery_pct, dtype: float64
```

Observation: - Annual income (r = 0.03): Higher-income borrowers do not consistently generate higher net recovery ratios. - DTI (r = 0.01): Debt-to-income ratio has almost no linear association with the recovery ratio. - Income vs. DTI (r = -0.13): A weak negative relationship exists between

these two borrower characteristics. These findings suggest that income and DTI alone are unlikely to explain much of the variation in recovery performance through a simple linear relationship.

Area Main finding

- Recovery occurrence: 67.64% of charged-off loans have positive recoveries
- Gross recovery: Average of 1172 per charged-off loan
- Net recovery: Average of 978 after reported collection fees
- Loan grade: Net recovery-to-loan ratio rises from 4.87% (A) to 7.91% (G)
- Loan term: 60-month loans average 6.75%, versus 5.70% for 36-month loans
- Origination year: Recent cohorts show lower recorded recoveries, potentially due to limited observation time
- Settlement completion: Completed settlements have a 21.86% average net recovery-to-loan ratio
- Annual income and DTI: Negligible linear correlations with net recovery ratio

[70]:
```
charged_off_df['loan_amount_group'] = pd.cut(␣
 ↪charged_off_df['loan_amnt'],bins=[0, 5000, 10000, 15000, 25000,␣
 ↪40000],labels=['0-5K', '5-10K', '10-15K', '15-25K', '25-40K'])

recovery_by_amount = charged_off_df.groupby('loan_amount_group',␣
 ↪observed=True)['net_recovery_pct'].agg(['count', 'mean', 'median']).round(2).
 ↪reset_index()

display(recovery_by_amount)
```

	loan_amount_group	count	mean	median
0	0-5K	28209	5.76	4.03
1	5-10K	62847	5.69	4.32
2	10-15K	57631	6.06	4.90
3	15-25K	76998	6.32	5.44
4	25-40K	36731	6.76	6.17

[71]:
```
fig = px.bar(recovery_by_amount,x='loan_amount_group', y='mean', title='Average␣
 ↪Net Recovery-to-Loan Ratio by Loan Amount')
fig.show()
```

[72]:
```
charged_off_df[['out_prncp', 'funded_amnt', 'total_rec_prncp', 'recoveries']].
 ↪describe()
```

[72]:

	out_prncp	funded_amnt	total_rec_prncp	recoveries
count	262,416.00	262,416.00	262,416.00	262,416.00
mean	0.00	15,522.52	4,364.35	1,172.25
std	0.00	8,793.96	4,101.54	1,823.16
min	0.00	500.00	0.00	0.00
25%	0.00	9,000.00	1,512.17	0.00
50%	0.00	14,225.00	3,141.40	549.37
75%	0.00	20,300.00	5,868.43	1,675.66
max	0.00	40,000.00	39,989.79	39,859.55

Key finding: Larger charged-off loans generally have higher net recovery-to-original-loan ratios. - Loans above $25K have the highest average ratio at 6.76%. - Loans between $5K and $10K have the lowest at 5.69%. - The median ratio also increases across the loan-size groups. However, this does not prove that larger loans are easier to recover.

0.0.6 Final business conclusions

1. Recovery occurrence declines across more recent origination cohorts. However, these cohorts had shorter observation periods, so differences in recorded recoveries cannot be interpreted as changes in collection effectiveness without adjusting for recovery time.
2. Loan grade and loan term are associated with differences in recovery outcomes. However, these differences may be affected by original loan sizes, remaining balances, and collection timelines.
3. Completed settlements are associated with substantially higher recorded recovery ratios. This is descriptive evidence, not proof that settlement completion causes improved recoveries.
4. Borrower income and DTI have weak linear associations with recovery ratios. Additional features or nonlinear analysis may be necessary to explain variations in collections.
5. Limitations to document: Our ratios use original funded amounts rather than balances outstanding at charge-off. We also lack confirmed charge-off and recovery transaction dates, so recovery timing cannot be consistently controlled. Settlement status is recorded after origination, and the reported collection fees may not capture all collection expenses
6. Loan size and recovery outcomes:Larger charged-off loans generally show higher net recovery-to-original-funded-amount ratios. Loans between $25,000 and 40,000 have an average ratio of 6.76%, compared with 5.69% for loans between 5,000 and 10,000. However, this relationship does not establish that larger loans are easier to recover, because borrower risk, collection timing, and unpaid balances may differ.

Vintage Analysis Vintage analysis groups loans by the period in which they originated and tracks their performance as they age.

[73]:
```
df['issue_month'] = df['issue_d'].dt.to_period('M').astype(str)

vintage_df = (df.groupby('issue_month').agg(total_loans=('loan_status',␣
 ↪'size'),charged_off=('is_charged_off', 'sum')).reset_index())

vintage_df['chargeoff_pct'] = (vintage_df['charged_off'] /␣
 ↪vintage_df['total_loans'] * 100).round(2)

vintage_df.head(10)
```

[73]:

	issue_month	total_loans	charged_off	chargeoff_pct
0	2007-06	24	3	12.50
1	2007-07	63	7	11.11
2	2007-08	74	20	27.03
3	2007-09	53	13	24.53

4	2007-10	105	34	32.38
5	2007-11	112	34	30.36
6	2007-12	172	47	27.33
7	2008-01	305	84	27.54
8	2008-02	306	61	19.93
9	2008-03	402	89	22.14

[74]:
```
vintage_df[['total_loans', 'chargeoff_pct']].describe()
```

[74]:

	total_loans	chargeoff_pct
count	139.00	139.00
mean	16,263.80	14.52
std	16,988.58	5.75
min	24.00	0.00
25%	950.50	12.73
50%	8,273.00	15.36
75%	33,805.00	17.37
max	61,992.00	32.38

[75]:
```
fig = px.line(vintage_df,x='issue_month',y='chargeoff_pct',title='Observed␣
↪Charge-Off Percentage by Origination Month',labels={'issue_month':␣
↪'Origination Month','chargeoff_pct': 'Charge-Off Percentage (%)'})

fig.show()
```

Observation:

- 2007–2008: Charge-offs fluctuate between 11–32%, highlighting the need to monitor smaller cohorts separately.
- 2009–2013: Observed charge-offs generally range from 10–19%, providing historical context for portfolio risk.
- 2014–2016: Relatively stable charge-offs of 15–19%, useful for identifying changes in lending performance.
- 2017–2018: Declining charge-offs largely reflect shorter observation periods, making direct comparisons misleading.

Business Impact: Cohort monitoring helps credit risk teams identify potential portfolio deterioration, investigate underwriting changes, and prioritize risk reviews. However, loan-age differences must be accounted for before making lending policy or loss forecasting decisions.

The Key issue with this dataset are:

1. Outcome maturity: Many 2017–2018 loans were still active when the dataset was collected.
2. Selection bias: Excluding active loans means we model resolved outcomes, not the complete lending population.
3. Missing charge-off dates: We cannot reliably define defaults within a fixed 12-, 24-, or 36-month horizon.

Model Feasibility Assessment is a preliminary check performed before building a predictive model to determine whether the available data can reliably support the business objective.

A model feasibility assessment was conducted to examine loan outcome maturity by origination year and term. Based on the results, the predictive model was restricted to resolved 36-month loans originated between 2007 and 2015, with chronological training, validation, and test periods.

[76]:
```
feasibility = pd.crosstab(
    [df['issue_d'].dt.year, df['term']],
    df['loan_status']
)[['Fully Paid', 'Charged Off', 'Current']]

feasibility['total'] = feasibility.sum(axis=1)

feasibility['unresolved_pct'] = (
    feasibility['Current'] / feasibility['total'] * 100
).round(2)

display(feasibility)
```

loan_status		Fully Paid	Charged Off	Current	total	unresolved_pct
issue_d	term					
2007	36	206	45	0	251	0.00
2008	36	1315	247	0	1562	0.00
2009	36	4122	594	0	4716	0.00
2010	36	7624	842	0	8466	0.00
	60	2425	645	0	3070	0.00
2011	36	12602	1499	0	14101	0.00
	60	5822	1798	0	7620	0.00
2012	36	37567	5903	0	43470	0.00
	60	7156	2741	0	9897	0.00
2013	36	88044	12378	0	100422	0.00
	60	25727	8644	9	34380	0.03
2014	36	140255	22315	0	162570	0.00
	60	40157	18741	13493	72391	18.64
2015	36	240805	42048	123	282976	0.04
	60	57332	33227	45070	135629	33.23
2016	36	170471	45075	102934	318480	32.32
	60	37561	21592	48984	108137	45.30
2017	36	96489	23943	191696	312128	61.42
	60	26032	12436	80068	118536	67.55
2018	36	30105	4260	303588	337953	89.83
	60	10135	2682	133730	146547	91.25

Observations: * 2007–2013: Historical cohorts are largely resolved, although we still need to account for other loan statuses. * 2014–2015: 36-month loans are almost entirely resolved, while many 60-month loans remain active. * 2016–2018: Large unresolved populations make simple Fully Paid versus Charged Off comparisons unsuitable for estimating default risk across all originated loans.

Model feasibility: Loan outcome maturity varies substantially by origination year and term. By 2015, nearly all 36-month loans had resolved outcomes in the primary status categories, whereas newer cohorts and 60-month loans had significant unresolved balances. The primary

model will therefore focus on historical 36-month originations, using chronological validation to reduce maturity-related bias.

I am selecting 36-month loans from 2007–2015 because they have had enough time to reach an outcome, allowing us to study charge-off risk more fairly. I would not include 60-month loans in the primary model because their outcome maturity differs substantially across the potential validation years. They can be considered in a separate analysis later.

[77]:
```
pd.crosstab(df.loc[(df['term'] == 36) & (df['issue_d'].dt.year <=␣
↪2015),'issue_d'].dt.year, df.loc[(df['term'] == 36) & (df['issue_d'].dt.year␣
↪<= 2015),'loan_status'])
```

[77]:

loan_status	Charged Off	Current
issue_d		
2007	45	0
2008	247	0
2009	594	0
2010	842	0
2011	1499	0
2012	5903	0
2013	12378	0
2014	22315	0
2015	42048	123

loan_status	Does not meet the credit policy. Status:Charged Off
issue_d	
2007	113
2008	249
2009	129
2010	158
2011	0
2012	0
2013	0
2014	0
2015	0

loan_status	Does not meet the credit policy. Status:Fully Paid	Fully Paid
issue_d		
2007	239	206
2008	582	1315
2009	436	4122
2010	532	7624
2011	0	12602
2012	0	37567
2013	0	88044
2014	0	140255
2015	0	240805

```
loan_status  In Grace Period  Late (16-30 days)  Late (31-120 days)
issue_d
2007                       0                  0                   0
2008                       0                  0                   0
2009                       0                  0                   0
2010                       0                  0                   0
2011                       0                  0                   0
2012                       0                  0                   0
2013                       0                  0                   0
2014                       0                  0                   0
2015                      15                  4                 178
```

Loan population: 36-month loans originated during 2007–2015.

Target variable: - 0 = Fully Paid, including special-policy Fully Paid - 1 = Charged Off, including special-policy Charged Off Exclude: Current, In Grace Period, and Late loans because their final outcomes are unknown.

Training: 2007–2013

Validation: 2014

Testing: 2015

[78]:
```
model_df = df[(df['term'] == 36) & (df['issue_d'].dt.year.between(2007, 2015))␣
↪& (df['loan_status'].str.contains('Fully Paid|Charged Off', na=False))].
↪copy()

model_df['target'] = (model_df['loan_status'].str.contains('Charged Off')).
↪astype(int)

model_df['issue_year'] = model_df['issue_d'].dt.year

model_df.groupby('issue_year')['target'].agg(['count', 'mean']).round(3).
↪reset_index()
```

[78]:
```
   issue_year   count  mean
0        2007     603  0.26
1        2008    2393  0.21
2        2009    5281  0.14
3        2010    9156  0.11
4        2011   14101  0.11
5        2012   43470  0.14
6        2013  100422  0.12
7        2014  162570  0.14
8        2015  282853  0.15
```

[79]:
```
features = [␣
↪'grade','loan_amnt','int_rate','annual_inc','dti','home_ownership','verification_status','p
```

[80]:
```
model_df[features].isna().mean().mul(100).round(2).sort_values(ascending=False)
```

[80]:
```
revol_util            0.06
grade                 0.00
loan_amnt             0.00
int_rate              0.00
annual_inc            0.00
dti                   0.00
home_ownership        0.00
verification_status   0.00
purpose               0.00
emp_length            0.00
delinq_2yrs           0.00
inq_last_6mths        0.00
dtype: float64
```

[81]:
```
X = model_df[features].copy()
y = model_df['target']

train = model_df['issue_year'] <= 2013
val = model_df['issue_year'] == 2014
test = model_df['issue_year'] == 2015

X_train, y_train = X.loc[train], y.loc[train]
X_val, y_val = X.loc[val], y.loc[val]
X_test, y_test = X.loc[test], y.loc[test]

print("Train:", X_train.shape)
print("Validation:", X_val.shape)
print("Test:", X_test.shape)
```

```
Train: (175426, 12)
Validation: (162570, 12)
Test: (282853, 12)
```

preprocessing - Median imputation for missing numerical values. - Standardization for numerical features. - One-hot encoding for categorical features. - Train-only capping of extreme numerical values.

[82]:
```
class OutlierCapper(BaseEstimator, TransformerMixin):
    def fit(self, X, y=None):
        self.upper_ = np.nanpercentile(X, 99.9, axis=0)
        return self

    def transform(self, X):
        return np.minimum(np.asarray(X, dtype=float), self.upper_)
```

[83]:
```
categorical = ['grade', 'home_ownership', 'verification_status','purpose',␣
 ↪'emp_length']

numerical = ['loan_amnt', 'int_rate', 'annual_inc', 'dti', 'delinq_2yrs',␣
 ↪'inq_last_6mths', 'revol_util']
```

[84]:
```
preprocessor = ColumnTransformer([
    ('cat', make_pipeline(
        SimpleImputer(strategy='most_frequent'),
        OneHotEncoder(handle_unknown='ignore', drop='first')
    ), categorical),

    ('num', make_pipeline(
        OutlierCapper(),
        SimpleImputer(strategy='median'),
        StandardScaler()
    ), numerical)
])
```

[85]:
```
model = make_pipeline(
    preprocessor,
    LogisticRegression(max_iter=1000)
)

model.fit(X_train, y_train)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/sklearn/linear_model/_logistic.py:451: OptimizeWarning:

Unknown solver options: iprint
```

[85]:
```
Pipeline(steps=[('columntransformer',
                 ColumnTransformer(transformers=[('cat',
Pipeline(steps=[('simpleimputer',
SimpleImputer(strategy='most_frequent')),
('onehotencoder',
OneHotEncoder(drop='first',
 handle_unknown='ignore'))]),
                                                 ['grade', 'home_ownership',
                                                  'verification_status',
                                                  'purpose', 'emp_length']),
                                                ('num',
Pipeline(steps=[('outliercapper',
OutlierCapper()),
('simpleimputer',
SimpleImputer(strategy='median')),
('standardscaler',
```

```
                                                  StandardScaler())]),
                                                                 ['loan_amnt', 'int_rate',
                                                                  'annual_inc', 'dti',
                                                                  'delinq_2yrs',
                                                                  'inq_last_6mths',
                                                                  'revol_util'])])),
                                ('logisticregression', LogisticRegression(max_iter=1000))])
```

[86]:
```
y_val_prob = model.predict_proba(X_val)[:, 1]

print("ROC-AUC:", round(roc_auc_score(y_val, y_val_prob), 3))
print("PR-AUC:", round(average_precision_score(y_val, y_val_prob), 3))
print("Charge-off prevalence:", round(y_val.mean(), 3))
```

```
ROC-AUC: 0.668
PR-AUC: 0.23
Charge-off prevalence: 0.137
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/sklearn/preprocessing/_encoders.py:246: UserWarning:

Found unknown categories in columns [1] during transform. These unknown
categories will be encoded as all zeros
```

Risk Decile Analysis

Banks use risk segmentation to examine whether higher model scores correspond to higher observed credit losses. We'll divide the 2014 validation loans into ten groups based on their predicted charge-off probabilities.

[87]:
```
risk_deciles = pd.DataFrame({
    'actual': y_val.to_numpy(),
    'predicted_prob': y_val_prob
})

risk_deciles['decile'] = pd.qcut(
    risk_deciles['predicted_prob'].rank(method='first'),
    10,
    labels=range(1, 11)
)

decile_summary = risk_deciles.groupby('decile', observed=True).agg(
    loans=('actual', 'size'),
    chargeoff_rate=('actual', 'mean'),
    avg_predicted_risk=('predicted_prob', 'mean')
)

decile_summary[['chargeoff_rate', 'avg_predicted_risk']] *= 100
```

```
display(decile_summary.round(2))
```

```
        loans  chargeoff_rate  avg_predicted_risk
decile
1       16257            4.31                3.84
2       16257            6.05                5.71
3       16257            8.11                7.42
4       16257            9.52                9.20
5       16257           11.64               10.79
6       16257           13.69               12.44
7       16257           15.89               14.31
8       16257           18.89               16.52
9       16257           21.63               19.48
10      16257           27.53               26.28
```

Observation: Risk Decile Analysis

- Observed charge-off rates increase consistently from 4.31% in the lowest-risk decile to 27.53% in the highest-risk decile, representing approximately 6.39 times higher observed charge-off risk. This demonstrates meaningful risk differentiation across the validation portfolio.
- Predicted probabilities generally underestimate observed charge-off rates, indicating potential calibration issues.

Business Impact: Risk segmentation can support portfolio monitoring and targeted credit-risk reviews. Further validation is required before using model scores for lending decisions.

The Brier score measures probability prediction error. lower is better, but it must be compared with a baseline to judge whether it's useful. Before selecting a classification threshold, we should check how closely predicted charge-off probabilities match actual outcomes.

[88]:
```
print("Brier Score:", round(brier_score_loss(y_val, y_val_prob), 4))
print("Average Predicted Risk:", round(y_val_prob.mean(), 4))
print("Actual Charge-Off Rate:", round(y_val.mean(), 4))
```

```
Brier Score: 0.1137
Average Predicted Risk: 0.126
Actual Charge-Off Rate: 0.1373
```

Probability Calibration — Key Insights

- Average predicted charge-off risk (12.61%) is below the observed rate (13.73%), indicating mild overall underprediction.
- The model achieved a Brier score of 0.1137 on the 2014 validation cohort, compared with approximately 0.1184 for a constant-probability baseline. However, the average predicted charge-off risk remained below the observed rate, indicating that probability calibration requires further assessment.

Business Impact: Better-calibrated probabilities can support more reliable credit-loss forecasting and risk-based portfolio monitoring.

Precsion and recall at different thresholds using the 2014 validation dataset

[89]:
```
results = []

for threshold in [0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50]:
    pred = (y_val_prob >= threshold).astype(int)

    results.append({
        'threshold': threshold,
        'precision': precision_score(y_val, pred, zero_division=0),
        'recall': recall_score(y_val, pred, zero_division=0),
        'f1': f1_score(y_val, pred, zero_division=0),
        'flagged_pct': pred.mean() * 100
    })

pd.DataFrame(results).round(3)
```

[89]:

	threshold	precision	recall	f1	flagged_pct
0	0.10	0.18	0.80	0.30	60.03
1	0.15	0.22	0.52	0.31	31.53
2	0.20	0.26	0.26	0.26	13.36
3	0.25	0.29	0.10	0.15	4.85
4	0.30	0.34	0.04	0.07	1.70
5	0.40	0.38	0.01	0.01	0.20
6	0.50	0.59	0.00	0.00	0.02

Classification Threshold Analysis — Key Insights

- 0.10 threshold: Captures 80% of charged-off loans but flags 60% of the portfolio.
- 0.15 threshold: Achieves the highest tested F1-score (0.31), with 52% recall and 31% of loans flagged.
- 0.20 threshold: Flags only 13% of loans, but recall falls to 26%.
- 0.50 threshold: Misses nearly all charged-off loans, making the default threshold unsuitable for charge-off detection.

Business Impact: Threshold selection involves balancing charge-off detection against false positives and review capacity. Risk-based segmentation is more appropriate until financial costs and operational constraints are established.

[90]:
```
pre = model.named_steps['columntransformer']

cat_names = pre.named_transformers_['cat'].named_steps[
    'onehotencoder'
].get_feature_names_out(categorical)

feature_names = list(cat_names) + numerical

coefficients = model.named_steps['logisticregression'].coef_[0]

coef_df = pd.DataFrame({
    'feature': feature_names,
```

```
    'coefficient': coefficients,
    'odds_ratio': np.exp(coefficients)
})

display(coef_df.sort_values('coefficient', ascending=False).round(3))
```

	feature	coefficient	odds_ratio
5	grade_G	0.97	2.63
4	grade_F	0.70	2.02
22	purpose_small_business	0.67	1.95
2	grade_D	0.65	1.92
3	grade_E	0.61	1.85
1	grade_C	0.56	1.74
14	purpose_educational	0.44	1.55
7	home_ownership_OTHER	0.43	1.54
35	emp_length_Unknown	0.43	1.53
0	grade_B	0.36	1.44
21	purpose_renewable_energy	0.23	1.26
37	int_rate	0.20	1.22
9	home_ownership_RENT	0.16	1.18
41	inq_last_6mths	0.14	1.15
18	purpose_medical	0.13	1.14
30	emp_length_6 years	0.11	1.12
6	home_ownership_NONE	0.11	1.11
39	dti	0.10	1.11
31	emp_length_7 years	0.09	1.10
36	loan_amnt	0.09	1.09
20	purpose_other	0.08	1.08
32	emp_length_8 years	0.07	1.07
15	purpose_home_improvement	0.06	1.06
42	revol_util	0.05	1.05
34	emp_length_< 1 year	0.05	1.05
8	home_ownership_OWN	0.04	1.04
33	emp_length_9 years	0.04	1.04
25	emp_length_10+ years	0.04	1.04
27	emp_length_3 years	0.04	1.04
29	emp_length_5 years	0.03	1.03
26	emp_length_2 years	0.03	1.03
19	purpose_moving	0.03	1.03
16	purpose_house	0.02	1.02
28	emp_length_4 years	0.02	1.02
10	verification_status_Source Verified	0.00	1.00
23	purpose_vacation	0.00	1.00
40	delinq_2yrs	-0.00	1.00
13	purpose_debt_consolidation	-0.05	0.95
11	verification_status_Verified	-0.08	0.92
17	purpose_major_purchase	-0.16	0.85
12	purpose_credit_card	-0.21	0.81

```
24                    purpose_wedding        -0.22        0.81
38                         annual_inc        -0.26        0.77
```

Coefficient Analysis

- Higher interest rates, debt-to-income ratios, and recent credit inquiries are associated with increased modeled charge-off risk.
- Higher annual income is associated with lower modeled charge-off risk.
- Loan grade and purpose contribute to risk differentiation, but individual categorical coefficients require cautious interpretation.
- Correlation between lender-assigned grades and interest rates may affect coefficient stability.

Business Impact: The model identifies borrower and loan characteristics associated with historical charge-offs, supporting risk segmentation and portfolio monitoring rather than standalone lending decisions.

[91]:
```
encoder = model.named_steps['columntransformer'].named_transformers_['cat'].
 ↪named_steps['onehotencoder']

print("Drop setting:", encoder.drop)
print("Total features:", model.named_steps['logisticregression'].coef_.shape[1])
```

```
Drop setting: first
Total features: 43
```

[92]:
```
from sklearn.metrics import roc_auc_score, average_precision_score,␣
 ↪brier_score_loss

y_test_prob = model.predict_proba(X_test)[:, 1]

print("Test Loans:", len(y_test))
print("Actual Charge-Off Rate:", round(y_test.mean() * 100, 2), "%")
print("ROC-AUC:", round(roc_auc_score(y_test, y_test_prob), 3))
print("PR-AUC:", round(average_precision_score(y_test, y_test_prob), 3))
print("Brier Score:", round(brier_score_loss(y_test, y_test_prob), 4))
print("Average Predicted Risk:", round(y_test_prob.mean() * 100, 2), "%")
```

```
Test Loans: 282853
Actual Charge-Off Rate: 14.87 %
ROC-AUC: 0.679
PR-AUC: 0.256
Brier Score: 0.1215
Average Predicted Risk: 11.6 %
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/sklearn/preprocessing/_encoders.py:246: UserWarning:

Found unknown categories in columns [1] during transform. These unknown
categories will be encoded as all zeros
```

[93]:
```
test_deciles = pd.DataFrame({
    'actual': y_test.to_numpy(),
    'predicted_prob': y_test_prob
})

test_deciles['decile'] = pd.qcut(
    test_deciles['predicted_prob'].rank(method='first'),
    10,
    labels=range(1, 11)
)

test_summary = test_deciles.groupby(
    'decile', observed=True
).agg(
    loans=('actual', 'size'),
    chargeoff_rate=('actual', 'mean'),
    avg_predicted_risk=('predicted_prob', 'mean')
)

test_summary[['chargeoff_rate', 'avg_predicted_risk']] *= 100

display(test_summary.round(2))
```

decile	loans	chargeoff_rate	avg_predicted_risk
1	28286	4.50	3.49
2	28285	5.87	5.23
3	28285	8.04	6.63
4	28285	10.40	8.24
5	28286	12.79	9.78
6	28285	14.42	11.37
7	28285	17.38	13.20
8	28285	20.56	15.39
9	28285	23.95	18.22
10	28286	30.75	24.49

[94]:
```
# Use the existing model's 2014 predicted probabilities

y_val_prob = model.predict_proba(X_val)[:, 1]

# Fit a simple calibration model
calibrator = LogisticRegression()

calibrator.fit(
    y_val_prob.reshape(-1, 1),
    y_val
)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/sklearn/preprocessing/_encoders.py:246: UserWarning:

Found unknown categories in columns [1] during transform. These unknown
categories will be encoded as all zeros

/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/sklearn/linear_model/_logistic.py:451: OptimizeWarning:

Unknown solver options: iprint
```

[94]:
```
LogisticRegression()
```

[95]:
```
# Original 2015 predictions

y_test_prob = model.predict_proba(X_test)[:, 1]

# Calibrated 2015 predictions
y_test_calibrated = calibrator.predict_proba(
    y_test_prob.reshape(-1, 1)
)[:, 1]
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/sklearn/preprocessing/_encoders.py:246: UserWarning:

Found unknown categories in columns [1] during transform. These unknown
categories will be encoded as all zeros
```

[96]:
```
comparison = pd.DataFrame({
    'Metric': [
        'ROC-AUC',
        'Brier Score',
        'Average Predicted Risk (%)',
        'Actual Charge-Off Rate (%)'
    ],
    'Original': [
        roc_auc_score(y_test, y_test_prob),
        brier_score_loss(y_test, y_test_prob),
        y_test_prob.mean() * 100,
        y_test.mean() * 100
    ],
    'Calibrated': [
        roc_auc_score(y_test, y_test_calibrated),
        brier_score_loss(y_test, y_test_calibrated),
        y_test_calibrated.mean() * 100,
        y_test.mean() * 100
```

```
    ]
})

display(comparison.round(4))
```

	Metric	Original	Calibrated
0	ROC-AUC	0.68	0.68
1	Brier Score	0.12	0.12
2	Average Predicted Risk (%)	11.60	12.75
3	Actual Charge-Off Rate (%)	14.87	14.87

[97]:
```
from sklearn.calibration import calibration_curve

actual_orig, pred_orig = calibration_curve(
    y_test, y_test_prob, n_bins=10, strategy='quantile'
)

actual_cal, pred_cal = calibration_curve(
    y_test, y_test_calibrated, n_bins=10, strategy='quantile'
)

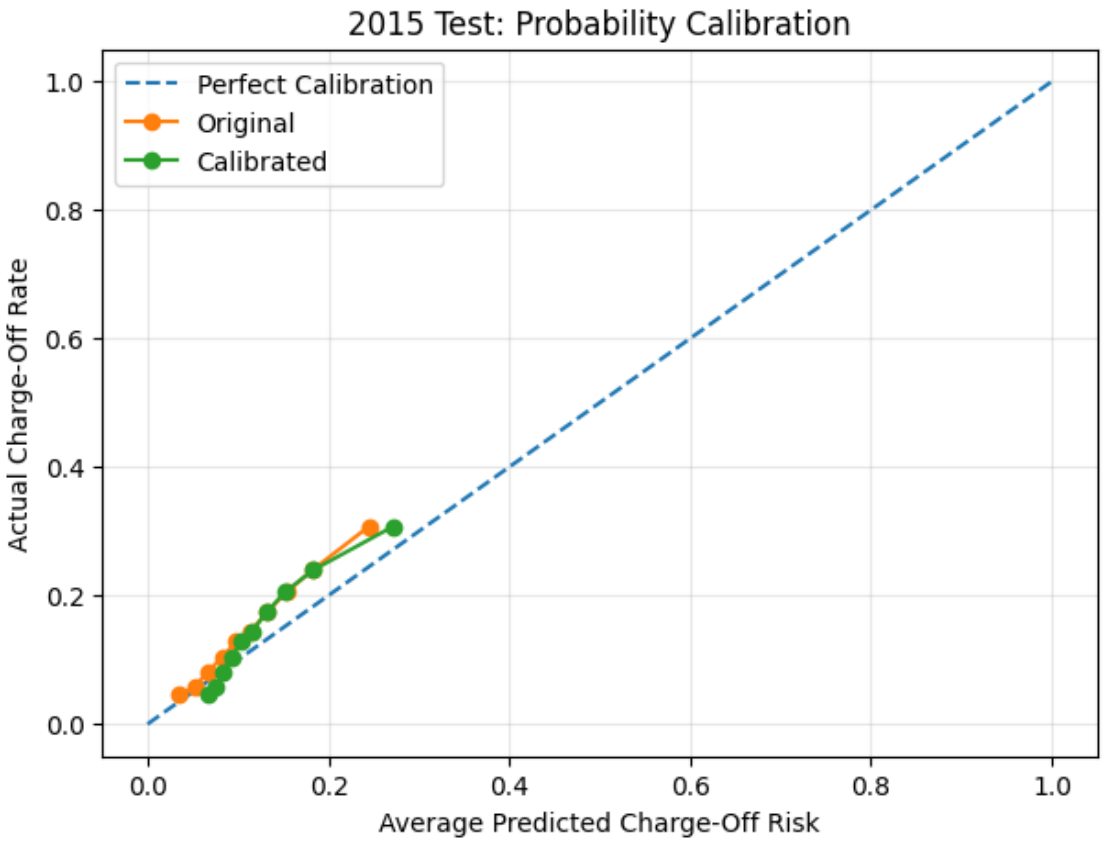
plt.figure(figsize=(7, 5))

plt.plot([0, 1], [0, 1], '--', label='Perfect Calibration')

plt.plot(
    pred_orig, actual_orig,
    marker='o', label='Original'
)

plt.plot(
    pred_cal, actual_cal,
    marker='o', label='Calibrated'
)

plt.xlabel('Average Predicted Charge-Off Risk')
plt.ylabel('Actual Charge-Off Rate')
plt.title('2015 Test: Probability Calibration')
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```



[98]:
```
print("Original Brier:",
      round(brier_score_loss(y_test, y_test_prob), 6))

print("Calibrated Brier:",
      round(brier_score_loss(y_test, y_test_calibrated), 6))
```

```
Original Brier: 0.121513
Calibrated Brier: 0.121378
```

Post-hoc Probability Recalibration — Key Findings

- Recalibration slightly improved the Brier score from **0.121513 to 0.121378**, indicating a small improvement in prediction accuracy.
- Average predicted charge-off risk increased from **11.60% to 12.75%**, moving closer to the actual rate of **14.87%**.
- ROC-AUC remained **0.679**, showing that the model's ability to rank risk did not change.
- **Conclusion:** Recalibration helped reduce risk underestimation, but the model is still better at identifying higher-risk loans than predicting exact charge-off probabilities.

Note: This was a post-hoc experiment using 2014 data for calibration and 2015 data for evaluation.

0.1 Final Credit Risk Model — Business Insights

- Achieved 0.679 ROC-AUC on 282,853 loans in the 2015 test cohort.
- Charge-off rates ranged from 4.50% to 30.75% across risk deciles, a 6.83× difference.
- The highest-risk 10% of loans accounted for 20.7% of total charge-offs.
- Recalibration slightly improved the Brier score (0.121513 to 0.121378), but predicted risk remained below actual charge-offs.

Business Implication: The model helps identify higher-risk loans for portfolio monitoring but requires further validation before use in lending decisions.

0.1.1 Model Limitations

- Selection bias: Only resolved 36-month loans from 2007–2015 were included, potentially excluding unresolved outcomes.
- Historical differences: LendingClub's underwriting and pricing policies may have changed over time.
- Lender-assigned features: Grade and interest rate reflect LendingClub's existing risk assessments.
- Calibration: Recalibration improved average predicted risk from 11.60% to 12.75%, but it remained below the actual 14.87% charge-off rate.
- Applicability: The model supports historical risk segmentation, not automated lending decisions or reliable individual default probability estimates.

[ ]: